

**DEPARTMENT OF INITIATIVES ORIENTED TO DEVELOPMENT AND INNOVATION
SUBDIRECTORATE FOR CENTERS AND ASSOCIATIVE RESEARCH**

TECHNOLOGY REGULAR RESEARCH TEAM PROJECTS 2024

RESEARCH PROPOSAL FORM

I. PROJECT TITLE (Same as in the online application system)

EASER: Evolution Assessment of Seismic Risk

II. IDENTIFICATION OF PARTICIPATING INSTITUTIONS

Main Institution	RUT
Universidad de Concepción	81.494.400-K

Interested Institution	RUT (if apply)
1.- Servicio Nacional de Prevención y Respuesta ante Desastres (SENAPRED)	60.509.001-K
2.- Ingeniería Estructural S.A.	99.514.240-6

Associate Institution	RUT
1.- Pontificia Universidad Católica de Chile	81.698.900-0
2.- Universidad de Chile	60.910.000-1

III. SCIENTIFIC SUMMARY/ RESUMEN CIENTÍFICO

Motivation and Problem Identification

Earthquakes are a major cause of human and economic losses (2.5 million fatalities, 4 to 10 million injuries, and economic losses surpassing \$2.7 trillion USD, 2023 adjusted). While preventing earthquakes is impossible, informed decisions can substantially mitigate seismic risk in earthquake-prone communities. Such decisions need a comprehensive evaluation of trade-offs, considering factors like retrofitting older structures, enforcing stringent building codes, and limiting urban plans. The imperative lies in understanding the resulting seismic risk of structures within a city, hinging on seismic hazard, structural vulnerability, and exposure. Over the past five decades, regional seismic risk assessment methodologies have proliferated, including HUD-funded reports, the ATC-13 project, HAZUS methodology, OpenQuake platform, the RPBEE approach, among others. However, these tools predominantly focus on stationary assessment, overlooking the dynamic interplay of hazard, vulnerability, and exposure over time. Our dynamic and fast-evolving world, with a projected global population of 9.77 billion by 2050 and a 68% urbanization rate, underscores the need for a time-dependent approach. The evolving exposure landscape, marked by the annual construction of taller urban structures, aligns with changing vulnerability due to aging, earthquake damage, and structural modifications. Similarly, the seismic hazard, typically uses stationary variables for recurrence, source characteristics, and site response during earthquakes, all of which evolve over time due to factors such as occurrences of large events, site consolidation due to global warming or vertical growth of the city, and advancements in scientific knowledge.

The project's response to this multifactorial challenge involves fostering interdisciplinary collaboration among Chilean universities, private institutions, government agencies, and international partners. EASER aims to develop a methodology for assessing the time evolution of seismic risk, focusing on residential buildings within the Andean seismotectonic environment. The resulting computational tools will empower stakeholders to make rational, time-dependent risk-management decisions, bridging the gap left by conventional stationary assessments and ensuring the resilience of assets in the face of evolving seismic hazards.

Objectives

Our goal is to assemble a cohesive, multidisciplinary group of researchers to advance the knowledge on the time dependency of seismic risk and to promote an environment that fosters collaboration and innovation, providing an international academic environment to train young earthquake scientists at the highest level. We will integrate seismology, geotechnical, and structural engineering to quantify the interrelation between seismic hazard time-dependency, seismic vulnerability time-dependency, and the exposure evolution over time, with epistemic and aleatory uncertainty considered in every step. Our results will ultimately allow the development of advanced computational tools to aid both public and private stakeholders in assessing more realistically the seismic risk in residential structures and communities within the Andean seismotectonic environment and making informed decisions to mitigate these risks. We will focus on four specific objectives (SO):

SO1: Develop time-dependent seismic hazard models within the Andean subduction context, considering the time between earthquakes, the temporal evolution of the source, path characteristics, and site response effects.

SO2: Evaluate the evolution of structural vulnerability over time due to degradation of materials and occurrence of earthquake sequences, both from analytical and experimental approaches.

SO3: Assess the temporal evolution of exposure through analysis of publicly available databases and image recognition techniques applied to multispectral satellite imagery.

SO4: Develop a set of computational tools to evaluate the effect of time-dependent seismic hazard, structural vulnerability and exposure on seismic risk, considering and propagating their uncertainties.

Hypothesis

The scientific hypothesis is that the seismic risk of structures and communities rapidly changes over time, strongly affected by changes in seismic hazard, seismic vulnerability, and exposure (e.g., population growth). Therefore, seismic risk needs to be studied considering its time-dependency. The main objective of our project is to investigate the specific impacts of each of these variables on the time evolution of seismic risk.

Concurrently, the technological hypothesis of this project is that both public and private stakeholders should consider the temporal evolution of seismic risk to make informed and rational decisions regarding risk mitigation strategies for future resilient communities.

Methodology

The project will be carried out through four main Research Lines (RLs), each one of them being directly related to one of the four SOs. The RL1 "Time evolution of seismic hazard" will integrate statistical and physical methods to develop time-dependent probabilistic models for seismic intensities. These models will integrate products of the temporal dimension, including data from paleoseismology, spatial variations of the crust's upper structure and segmentation, and shallow crustal seismogenic faults. As a result of this, it will be possible to quantitatively account for the time-dependent probability that an earthquake of a given magnitude occurs in a particular region, with surface intensities that consider several aspects of the physical process that are currently neglected (i.e., considered uncertainties). This research line provides valuable inputs for RL2 and RL4. The RL2 "Time evolution of seismic vulnerability" will characterize changes in structural vulnerability through analytical time-dependent fragility functions, particularly Chilean building typologies of reinforced concrete and masonry. The team will evaluate key-factors that influence the vulnerability in time within the Chilean context, such as cumulative earthquake damage and chloride- and carbonation-induced corrosion will be evaluated. The



research line involves selecting archetype buildings representative of prevalent building typologies, developing nonlinear analytical models with time-dependent effects, and validating them using experimental data collected from ambient vibration tests on existing structures. Analytical fragility functions will be developed for selected buildings, accounting for time-dependent effects. The temporal evolution of fragility functions for each archetype will be analyzed, and correlations will be established between time-dependent and time-independent fragility functions across different building types. The RL3 “Time evolution of exposure” will investigate the temporal evolution of urban exposure, focused on residential structures in the cities of Santiago, Viña del Mar, and Concepción. It will encompass aspects such as the number of residential structures, their locations, and various properties, including age since construction, building height, among others. This will be done by processing historical aerophotogrammetry and satellite image data, and also publicly available building data gathered from repositories of governmental entities. Also, mobile cellphone data will be integrated into the process. Later, an enhanced time-dependent exposure model will be developed, with the objective capability of forecasting exposure 30 years in the future. Finally, the developed model will be subjected to a validation experiment employing historical data. The RL 4 “Development of computational tools for evaluating time-dependent seismic risks” will focus on the technological aspect of the project, which has two primary tasks. The first one involves the development of a set of tools for the time-dependent evaluation of seismic risk, entitled EASER model. This RL will integrate the outcomes from the other research lines. These tools will be curated and made accessible through public repositories. The second task entails using these tools to compare future seismic risk mitigation strategies in the cities of Santiago, Viña del Mar, and Concepción. These three cities are selected because of their significance for Chile in terms of both population and economy. Several risk mitigation strategies will be evaluated through the EASER models.

The project will be carried out by robust team of researchers from the three of the best Chilean universities (Universidad de Concepción, Universidad de Chile, and Pontificia Universidad Católica de Chile), prestigious international collaborators from 6 different countries (U.S., Germany, Italy, England, Austria, and Colombia) and also from Chile, a group of 10 students of different levels (postdoctorate, doctorate, and masters’), and other collaborators with smaller but equally significant participations. The team will work in collaboration with two interested institutions: SENAPRED (Chilean National Service for Disaster Prevention and Response) and VMB engineering consultancy. A continuous transfer of information and feedback with these institutions will enable the team to generate results that effectively have value for society and stakeholders. Furthermore, dissemination of results and gender strategy are considered, which add value to the outcomes of the project. Thus, huge human resources are involved, which almost ensures the success of the project.

Scientific and Technological Merits

The scientific merit of EASER lies in unraveling the intricate effects of seismic hazard, vulnerability, and exposure evolution on the seismic risk of urban regions. Some of the questions that will be explored are: Which factor—seismic hazard, vulnerability, or exposure—exerts a greater influence on seismic risk evolution? After a large event where vulnerability increases due to earthquake-induced damage, but the time-dependent hazard decreases, what are the resultant implications for seismic risk? How do cities grow, and how does this growth impact their seismic risk? The EASER project will result in an integrated methodology to systematically incorporate the time dependency of seismic hazard, vulnerability, and exposure, enhancing our understanding of seismic risk’s dynamic nature.

The technological merit of the project resides in the EASER model, offering a comprehensive framework for assessing seismic risk while considering the temporal evolution of its elements. This model stands as a rational and transparent tool accessible to both public and private stakeholders, facilitating the evaluation of diverse seismic risk mitigation strategies. Furthermore, the EASER model will empower stakeholders to make informed decisions, contributing to the development of more resilient cities.

Extension and Dissemination of Results

The topics related to seismic risk are comprehensively addressed, and dissemination activities reach professionals and the non-specialized public. The knowledge transfer and dissemination activities will be divided into two large groups: (1) a non-specialized and/or scientific public and (2) a specialized and/or scientific public. The results will be communicated to the government entities responsible for seismic hazards in Chile (SENAPRED, SHOA, SERNAGEOMIN, CSN), starting with our partner SENAPRED, with whom we have the agreement to work closely. The planned activities include reels for social media, podcasts, joining STEM promotion groups at school levels, seminars to promote project results among national professional societies, national and international conferences, and collaborative journal publications.

Motivación e identificación del problema

Los terremotos son una de las principales causas de pérdidas humanas y económicas (2,5 millones de víctimas, entre 4 y 10 millones de heridos, y pérdidas económicas que superan los 2,7 billones de dólares, ajustados a 2023). Aunque es imposible prevenir los terremotos, las decisiones bien fundadas pueden mitigar sustancialmente el riesgo sísmico en las comunidades propensas a sufrirlos. La clave está en comprender el riesgo sísmico de una ciudad, en función de la amenaza sísmica, la vulnerabilidad estructural y la exposición.



En las últimas décadas han proliferado las metodologías de evaluación del riesgo sísmico regional, incluidos los informes financiados por el HUD, el proyecto ATC-13, la metodología HAZUS, la plataforma OpenQuake, el enfoque RPBE, entre otros. Sin embargo, estas herramientas se centran en una evaluación estacionaria, pasando por alto la interacción dinámica de la amenaza, vulnerabilidad y exposición en el tiempo. Sin embargo, nuestro mundo dinámico y en rápida evolución, con una población mundial prevista de 9.770 millones de personas para 2050 y una tasa de urbanización del 68%, enfatiza la necesidad de un enfoque dependiente del tiempo. El cambiante panorama de la exposición, marcado por la construcción anual de más estructuras urbanas y más altas, se alinea con la vulnerabilidad cambiante producto del envejecimiento, los daños sísmicos y las modificaciones estructurales. Del mismo modo, la amenaza sísmica suele utilizar variables estacionarias para la recurrencia, las características de la fuente y la respuesta del emplazamiento durante los terremotos, todas las cuales evolucionan con el tiempo debido a factores como la ocurrencia de grandes eventos, la consolidación del emplazamiento debido al calentamiento global o al crecimiento vertical de la ciudad, y los avances en el conocimiento científico.

En respuesta a este reto multifactorial, este proyecto fomenta la colaboración interdisciplinaria entre universidades, instituciones privadas, organismos gubernamentales y socios internacionales. EASER busca desarrollar una metodología y herramientas computacionales para evaluar el riesgo sísmico incorporando la variación temporal de los tres elementos clave. Nos enfocamos en edificaciones residenciales en ciudades de Chile, para proporcionar al sector público y privado herramientas para evaluar de forma realista el riesgo sísmico en función del tiempo. Ello que permitirá tomar decisiones informadas para reducir eficazmente estos riesgos, llenando la brecha que dejan las evaluaciones estacionarias convencionales y garantizando la resiliencia frente a la evolución de los peligros sísmicos.

Objetivos

Nuestro objetivo es reunir un grupo cohesionado y multidisciplinario de investigadores para avanzar en el conocimiento de la dependencia temporal del riesgo sísmico y promover un ambiente que fomente la colaboración y la innovación, proporcionando un entorno académico internacional para formar a jóvenes científicos especializados en terremotos al más alto nivel. Integraremos sismología, geotecnia e ingeniería estructural para cuantificar las interrelaciones entre la amenaza sísmica, la vulnerabilidad sísmica y la evolución de la exposición a lo largo del tiempo, teniendo en cuenta la incertidumbre epistémica y aleatoria en cada paso. Nuestros resultados permitirán el desarrollo de herramientas computacionales avanzadas que permitan a las partes interesadas, tanto públicas como privadas, evaluar de forma más realista el riesgo sísmico en estructuras residenciales y comunidades dentro del entorno sismotectónico andino y a tomar decisiones informadas para mitigar estos riesgos. Nos centraremos en cuatro objetivos específicos (OE):

OE1: Desarrollar modelos de amenaza sísmica dependientes del tiempo dentro del contexto de subducción andino, considerando el tiempo entre terremotos, la evolución temporal de los grados de bloqueo en la fuente, las características de la trayectoria y los efectos de respuesta del sitio.

OE2: Evaluar la evolución de la vulnerabilidad estructural en el tiempo debido a la degradación de los materiales y la ocurrencia de secuencias sísmicas, tanto desde enfoques analíticos como experimentales.

OE3: Evaluar la evolución temporal de la exposición mediante el análisis de bases de datos públicas y técnicas de reconocimiento de imágenes aplicadas a imágenes de satélite multispectrales.

OE4: Desarrollar un conjunto de herramientas computacionales para evaluar el efecto de la peligrosidad sísmica dependiente del tiempo, la vulnerabilidad estructural y la exposición en el riesgo sísmico, considerando y propagando sus incertidumbres.

Hipótesis

La hipótesis científica es que el riesgo sísmico varía con el tiempo, fuertemente afectado por los cambios en la amenaza, la vulnerabilidad y la exposición (por ejemplo, el crecimiento de la población). Por lo tanto, el riesgo sísmico debe estudiarse teniendo en cuenta su dependencia temporal. El principal objetivo de nuestro proyecto es investigar los impactos específicos de cada una de estas variables en la evolución temporal del riesgo sísmico.

A la vez, la hipótesis tecnológica del proyecto es que las partes interesadas, públicas y privadas, deben considerar la evolución temporal del riesgo sísmico para tomar decisiones informadas y racionales en relación con las estrategias de mitigación del riesgo para futuras comunidades resilientes.

Metodología

El proyecto se llevará a cabo a través de cuatro líneas de investigación principales (LI), cada una directamente relacionada con uno de los OE.

La LI1 "Evolución temporal de la amenaza sísmica" integrará métodos estadísticos y físicos para desarrollar modelos probabilísticos dependientes del tiempo para distintas intensidades sísmicas. Estos modelos integrarán productos de dimensión temporal, incluyendo datos de paleosismología, variaciones espaciales de la estructura superficial y segmentación de la corteza, y fallas sismogénicas superficiales de la corteza. Como resultado, se podrá cuantificar la probabilidad dependiente del tiempo de un terremoto de magnitud dada en una región en particular, con intensidades superficiales que consideran varios aspectos del proceso físico ignorados (i.e., incertidumbres consideradas). Esta línea de investigación aporta valiosos insumos para LI2 y LI4. La LI2 "Evolución temporal de la vulnerabilidad sísmica" desarrollará funciones analíticas de fragilidad dependientes del tiempo,



especialmente en tipologías estructurales chilenas de hormigón armado y mampostería. Se evaluarán factores clave que influyen en la vulnerabilidad en el tiempo, como los daños acumulados por terremotos y la corrosión inducida por cloruros y carbonataciones. Se seleccionarán edificios arquetipo representativos de tipologías de edificios prevalentes, desarrollaremos modelos analíticos no lineales con efectos dependientes del tiempo, los que serán validados usando datos experimentales de pruebas de vibración ambiental en estructuras existentes. Se desarrollarán funciones analíticas de fragilidad teniendo en cuenta los efectos dependientes del tiempo. Se analizará la evolución temporal de las funciones de fragilidad, y se establecerán correlaciones entre las funciones de fragilidad dependientes e independientes del tiempo en diferentes tipos de edificios. La LI3 "Evolución temporal de la exposición" investigará la evolución temporal de la exposición urbana, con foco en estructuras residenciales de las ciudades de Santiago, Viña del Mar y Concepción. Abarcará aspectos tales como el número de estructuras residenciales, sus ubicaciones y diversas propiedades como la antigüedad y la altura del edificio, entre otros. Esto se hará mediante el procesamiento de datos históricos de aerofotogrametría e imágenes satelitales, y también datos de edificios disponibles en bases de datos públicas. Además, datos de teléfonos móviles se integrarán en el proceso. A partir de esta información se desarrollará un modelo de exposición dependiente del tiempo, para pronosticar la exposición futura a 30 años, que se validará con datos históricos. La LI 4 "Desarrollo de herramientas computacionales para la evaluación de riesgos sísmicos dependientes del tiempo" se centrará en el aspecto tecnológico del proyecto, la que tiene dos tareas principales. La primera consiste en el desarrollo de un conjunto de herramientas para la evaluación del riesgo sísmico dependiente del tiempo, denominado modelo EASER, integrando los resultados de las otras líneas de investigación. Estas herramientas serán accesibles a través de repositorios públicos. La segunda tarea consiste en utilizar estas herramientas para comparar futuras estrategias de mitigación del riesgo sísmico en las ciudades de Santiago, Viña del Mar y Concepción. Estas tres ciudades son seleccionadas por su importancia para Chile, en población y economía. Se evaluarán varias estrategias de mitigación de riesgos a través de los modelos EASER.

El proyecto será llevado a cabo por un robusto equipo de investigadores de las tres mejores universidades chilenas (Universidad de Concepción, Universidad de Chile y Pontificia Universidad Católica de Chile), prestigiosos colaboradores internacionales de 6 países diferentes (EE.UU., Alemania, Italia, Inglaterra, Austria y Colombia) y también de Chile, un grupo de 10 estudiantes de diferentes niveles (postdoctorado, doctorado, y maestría), y otros colaboradores con participaciones más pequeñas pero igualmente significativas. El equipo trabajará en colaboración con dos instituciones interesadas: SENAPRED (Servicio Nacional de Prevención y Atención de Desastres) y VMB Consultoría de Ingeniería. Una transferencia continua de información y retroalimentación con estas instituciones permitirá al equipo generar resultados que efectivamente tengan valor para la sociedad y los grupos de interés. Además, se considera la difusión de los resultados y una estrategia de género, que agregan valor a los resultados del proyecto. Por lo tanto, se involucran enormes recursos humanos, lo que casi asegura el éxito del proyecto.

Méritos Científicos y Tecnológicos

El mérito científico de EASER radica en entender los complejos efectos de la amenaza, la vulnerabilidad y la evolución de la exposición en el riesgo sísmico de zonas urbanas. Algunas de las preguntas que se explorarán son: ¿Qué factor (amenaza, vulnerabilidad o exposición) ejerce una mayor influencia en la evolución del riesgo sísmico? Tras un gran evento en el que la vulnerabilidad aumenta por los daños causados por los terremotos, pero el peligro dependiente del tiempo disminuye, ¿qué implicaciones resultan en el riesgo sísmico? ¿Cómo el crecimiento de las ciudades impacta en su riesgo sísmico? El proyecto EASER producirá una metodología integrada para incorporar sistemáticamente la dependencia temporal de la amenaza, la vulnerabilidad y la exposición, lo que mejorará nuestra comprensión de la naturaleza dinámica del riesgo sísmico.

El mérito tecnológico del proyecto reside en el modelo EASER, el cual ofrece un marco integral para evaluar el riesgo sísmico considerando la evolución temporal de sus elementos. Este modelo constituirá una herramienta racional y transparente, accesible tanto a los actores públicos como privados, facilitando la evaluación de diversas estrategias de mitigación del riesgo sísmico.

Extensión y difusión de resultados

Los temas relacionados con el riesgo sísmico son abordados de manera integral, y las actividades de difusión llegan a profesionales y público no especializado. Las actividades de transferencia y difusión del conocimiento se dividirán en dos grandes grupos: (1) un público no especializado y/o científico y (2) un público especializado y/o científico. Los resultados serán comunicados a las entidades gubernamentales responsables de la amenaza sísmica en Chile (SENAPRED, SHOA, SERNAGEOMIN, CSN), comenzando por nuestro socio SENAPRED con quien tenemos el acuerdo de trabajar estrechamente. Las actividades previstas incluyen reels para las redes sociales, podcasts, unirse a grupos de promoción de STEM a nivel escolar, seminarios para promover los resultados de los proyectos entre las sociedades profesionales nacionales, conferencias nacionales e internacionales y publicaciones en revistas colaborativas.



IV. PROJECT DESCRIPTION

a. Research and development cycle

1. Motivation and Problem Identification

Since 1900, global earthquakes have caused approximately 2.5 million fatalities, 4 to 10 million injuries, and over USD 2.7 trillion (2023 adjusted USD) in economic losses, equivalent to an average annual economic loss of approximately \$22 billion USD¹. Although it is impossible to prevent earthquakes from happening, we can make informed decisions to mitigate the seismic risk of communities located in earthquake-prone regions. These decisions involve carefully evaluating trade-offs between risk mitigation costs (e.g., mandating retrofitting plans for older structures, enforcing stricter design criteria for new structures, or limiting building areas) and the reduction of potential negative consequences from future earthquakes. Therefore, understanding our structures' and communities' resulting seismic risk is crucial for making future rational risk-management decisions (e.g., urban planning, seismic design codes development, earthquake insurance quotation, or site improvements). The seismic risk of structures within a city depends on three key elements: seismic hazard, structural vulnerability, and exposure (Figure 1). In the context of this project, "seismic hazard" describes the likelihood of having various ground-shaking intensities in a specific location due to earthquakes in the surrounding region; "structural vulnerability" characterizes the probability that a certain structure will be damaged when subjected to an earthquake ground motion; and "exposure" refers to the geolocation and attributes of the structures within the city that is exposed to the hazard. Over the past 50 years, several methodologies have been developed for evaluating the seismic risk of cities on a regional scale, including the HUD-funded reports²⁻¹², the ATC-13 project¹³; the HAZUS methodology¹⁴⁻¹⁶, the OpenQuake platform¹⁷, and the RPBE approach¹⁸, among several others. However, these tools primarily focus on a stationary assessment of the seismic risk; in other words, they evaluate the seismic risk considering a single time point in the past (or present) without projecting the change in hazard, vulnerability, and exposure into the future.

We live in a dynamic and constantly evolving world: whereas the current global population exceeds 8 billion (double that of 50 years ago), by 2050 it is projected to reach 9.77 billion, with a 68% urbanization rate^{19,20}. This dynamic landscape entails rapidly changing exposure over time, marked by the annual construction of taller and more residential structures in urban areas. Concurrently, the vulnerability of structures evolves due to factors such as aging, previous earthquake damage and structural modifications. For example, lack of periodic maintenance can lead to the degradation of structural strength over time^{e.g., 21}. Modifications made to structures also modify their vulnerability; for instance, informal construction, common in many regions of the world, usually deviates from adequate design and construction standards, significantly diminishing the structural capacity^{e.g., 22}. Additionally, frequent seismic events in earthquake-prone regions may impact structural vulnerability due to nonlinear incursions^{e.g., 23,24}. Structures that experience slight or no damage during seismic events are often not restored to their original configuration, resulting in altered vulnerability for subsequent earthquakes conditioned on their prior damage^{e.g., 25}.

The seismic hazard also evolves over time. Although a memoryless (i.e., time-independent) Poisson process is typically used to characterize earthquake occurrences within a region, both models and empirical observations suggest a departure from that assumption^{e.g., 26,27}. A memoryless model assigns a constant probability of occurrence of an earthquake, independent of the time since the last event (i.e., the probability of a magnitude 8 earthquake is the same the day before and the day after the occurrence of another magnitude 8 event). However, large seismic events require the accumulation of high stresses before nucleation, indicating that the probability of a significant earthquake depends on the time since the last event, leading to an increase in seismic hazard with time since the last occurrence. Even these time-dependent models (e.g., Brownian Passage Time, Weibull) are currently calibrated on existing data that we know will evolve with more knowledge, this associated epistemic uncertainty needs to be accounted for if we want to fully consider the time dependency of seismic risk. The evolution of knowledge can also significantly impact seismic hazard assessments. For instance, in Santiago, Chile, a west-dipping reverse shallow crustal fault, known as the San Ramón Fault (SRF), capable of triggering earthquakes with magnitudes exceeding 7.0, was confirmed as active in 2002²⁸. Results from this discovery and posterior research²⁹⁻³² lead to a new seismic source to be considered for seismic hazard studies in Santiago. Seismic hazard also evolves due to changes in the coupling degree at the source^{e.g., 33,34}, which can happen due to nearby earthquakes or gradually due to plates relative movement.

Site conditions also affect seismic hazard significantly, and the evolution of these effects has been documented^{e.g., 35,36}, yet these variations are not considered in the future response of, for example, alluvial basins. This variability of site conditions can, and should, be mapped into seismic hazard. Another site condition responsible for most earthquake-related damage, in the absence of a great tsunami, is seismic-induced liquefaction. Liquefaction is triggered by earthquake shaking but requires a liquefaction-prone site to occur. Consequences of liquefaction include settlements and lateral spreading, which cause damage to all kinds of civil infrastructure. The triggering of liquefaction varies from the well-studied cases of shallow crustal earthquakes to the much less-studied liquefaction due to megathrust events^{37,38}. A probabilistic liquefaction hazard analysis includes the probability of an earthquake occurrence, the likelihood of liquefaction triggering, the likelihood of surface manifestation (e.g., settlements), and the likelihood of these causing damage to a structure. Yet, all of these calculations consider stationary source, path, and site characteristics³⁹.

Considering the described rapid evolution of exposure, vulnerability, and seismic hazard (Figure 1), a stationary assessment of the seismic risk, as commonly practiced when risk is evaluated, undermines the opportunity to make long-term rational mitigation decisions.

In response to this challenge, this project will foster an interdisciplinary collaboration among public and private Chilean universities, private institutions, government agencies, and international research partners to develop a methodology for assessing the seismic risk of cities, considering the time evolution of its three key elements. In particular, the focus will be on residential buildings, due to their significant social impact, and the Andean seismotectonic environment. The resulting methodology will be then packaged into a set of computational tools, enabling both public and private stakeholders to evaluate the time-dependent seismic risk of their assets and make rational risk-management decisions for the future.

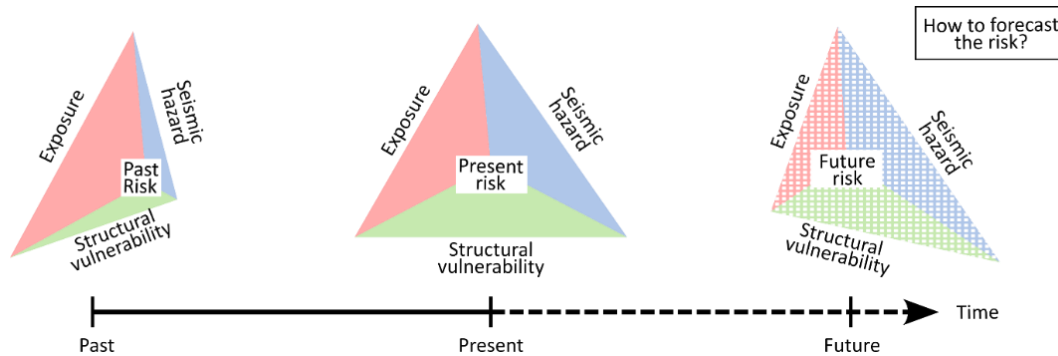


Figure 1. Elements of seismic risk and their evolution over time.

2. State of the Art

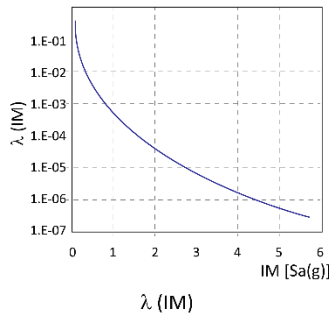
2.1. Stationary assessment of seismic risk

Several methodologies for seismic risk assessment of individual structures (i.e., "structure-specific") were proposed during the 1980s and 1990s^{40–42}. The M_w 6.9 1989 Loma Prieta and the M_w 6.7 1994 Northridge earthquakes demonstrated that even moderate-magnitude events occurring near large metropolitan areas can result in large economic losses, extended downtimes, thousands of people displaced from their homes, and several fatalities. The unexpected large consequences of these two moderate-magnitude seismic events triggered several early efforts in what is referred to as "performance-based earthquake engineering" (PBEE). The First Generation of PBEE^{e.g., 43,44} focused on assisting stakeholders on making rational risk management decisions. This is how researchers started to explicitly establish performance levels, based on component deformation demands rather than forces, to be achieved at different ground-motion intensity levels. However, that first generation of PBEE was primarily based on deterministic approaches and on the evaluation of performance at the component level. During the early 2000s, PBEE finally evolved into a rigorous probabilistic framework that was formalized by the Pacific Earthquake Engineering Research (PEER) Center in what is commonly referred to as the PEER PBEE framework^{45–48}. In this framework, the seismic risk assessment procedure is based on the explicit evaluation and propagation of the uncertainties of four types of random variables, which are assessed separately at four different analysis stages (Figure 2): (1) Intensity Measure, IM, characterizing the probability of exceedance of the ground motion intensity at a given site through Probabilistic Seismic Hazard Analysis (PSHA)^{49,50}; (2) Engineering Demand Parameter, EDP, describing the probability distribution of the structural response conditioned on the level of IM; (3) Damage Measure, DM, describing various levels of physical damage in structural and nonstructural elements through fragility curves that provide the probability of damage conditioned on EDP; and (4) Decision Variable, DV, quantifying the consequences of seismic damage (i.e., economic losses, downtime, and casualties) through loss curves that provide the probability distribution of DV conditioned on DM.

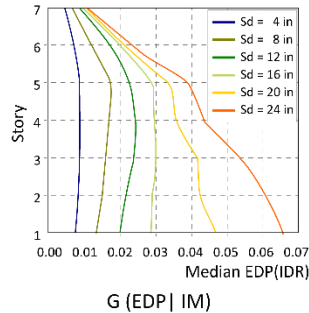
The PEER PBEE framework represents the state-of-the-art in seismic risk evaluation. It serves the purpose of providing estimates of the seismic risk of a structure in terms of relevant metrics for stakeholders. For example, it can answer questions such as, among others: (1) What is the probability of collapse of a particular building given a specific earthquake scenario?; (2) What is the probability of collapse of a particular structure in 50 years considering all possible earthquakes in the surrounding region that may affect the site?; (3) What is the expected number of damaged residential units in a particular building given an earthquake scenario?; and (4) What is the expected annual loss of a particular structure considering every possible earthquake in the surrounding region?

(A) Intensity Measure (IM)**Probabilistic Seismic Hazard Analysis (PSHA)**

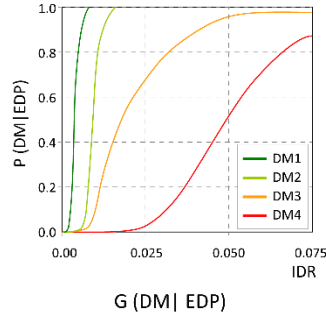
PGA, PGV, Sa(T), IA, duration

**(B) Engineering Demand Parameter (EDP)****Structural Analysis**

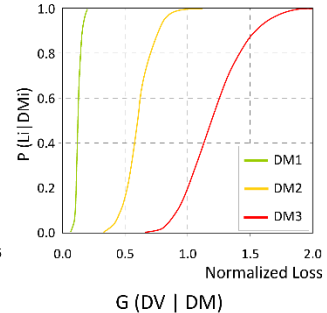
IDR, PFA, RIDR

**(C) Damage Measure (DM)****Damage Analysis**

Slight, Moderate, Extensive, Collapse

**(D) Decision Variable (DV)****Loss Analysis**

Economic Losses, Downtime, Casualties

**Figure 2.** Components of the PEER PBEE framework for structure-specific seismic performance assessment (source: E. Miranda).

Although structure-specific risk assessment methodologies are aimed at giving accurate predictions for individual structures, they cannot be easily extrapolated to a regional scale, mainly due to two reasons:

1. Structure-specific methodologies require the use of detailed structural models and nonlinear response history analyses. However, at present time it is computationally unfeasible to use these detailed models and complex analyses in a large number of structures (e.g., thousands) simultaneously. Therefore, regional seismic risk assessment studies must rely on simplified models or precomputed fragility curves to estimate the seismic performance of individual structures.
2. When evaluating the seismic performance of a group of structures, there are several correlations that need to be explicitly considered, both between seismic demands at different sites^{51–56}, and between structural capacities of different structures^{57,58}. Since these correlations tend to be positive, if they are neglected, the uncertainty of aggregated performance metrics (e.g., the number of damaged buildings, or the total economic losses within the region) may be significantly underestimated. Consequently, the probabilities of large-consequence events that are of utmost importance for risk management decisions (e.g., an earthquake resulting in a large number of collapsed structures) are significantly underestimated⁵⁸.

Until the 1970s, regional seismic risk assessments were mostly a concern of the insurance industry⁵⁹. In the late 1960s and the 1970s, the Office of the Secretary of the U.S. Department of Housing and Urban Development (HUD) supported a series of 11 studies to document and evaluate seismic damage and losses in different types of buildings in various earthquakes. These constitute the earliest government-supported studies on regional seismic risk assessment in USA^{2–12}. In these reports, the authors used deterministic relationships between Modified Mercalli Intensity (MMI) and damage in different types of structures. In 1973, Whitman et al.⁶⁰ proposed the concept of Damage Probability Matrices (DPM), which provide the probability that a structure experiences a given level of damage conditioned on the MMI at the site. Later in 1982, the Applied Technology Council (ATC) developed a methodology for evaluating seismic damage and losses in California, commonly known as ATC-13¹³. Authors of this report proposed DPMs based on expert opinion for 78 classes of facilities, including 40 building types and 38 lifeline structures.

After a series of studies focused on providing guidelines for a future standardized seismic risk assessment methodology for the USA^{61,62}, HAZUS was presented in 1997^{14–16}. HAZUS is a Geographic Information System (GIS)-based software for risk estimation that is periodically reviewed and is the most commonly used approach for regional seismic loss estimation by federal, state, and local governments in USA. Unlike previous methodologies that use MMI for characterizing seismic intensities, HAZUS uses objective parameters such as peak ground acceleration, PGA, and spectral acceleration ordinates, Sa(T). This represents a significant advantage, as it is difficult to correctly estimate the level of damage in specific structures as a function of the general level of damage provided by MMI, which is itself a measure of damage (in other words, MMI is estimated from the general damage in the region, and then we try to estimate the damage in specific structures from the MMI; so damage is estimated as a function of damage, which is clearly and conceptually incorrect). Despite the use of PGA and Sa(T) as measures of ground-motion intensity, the HAZUS methodology still has many drawbacks. In particular, it only considers mean estimates of ground-motion intensities given an earthquake scenario, neglecting their variability and spatial correlations. Moreover, the structural response given the ground-motion intensity is estimated using equivalent linearization and capacity spectrum procedures, which have been demonstrated to have problems such as resulting in significantly biased estimates, and having multiple solutions or lack of convergence^{63,64}. Finally, fragility curves in HAZUS were developed from expert judgment and calibrated to previous earthquakes. Thus, it has been questioned to be overfitted to those previous events⁶⁵.

Approximately fifteen years ago, the non-profit Global Earthquake Model (GEM) Foundation was created with the main goal of creating a model of the seismic hazard and risk of the entire world for risk management decision-making. In 2013, this effort led to an open-source platform for seismic hazard and seismic risk modeling, entitled OpenQuake¹⁷. This platform is constantly revised and



updated by researchers around the world, and incorporates the state-of-the-art in regional seismic risk assessment. For example, unlike HAZUS, OpenQuake explicitly considers the variability and spatial correlation of ground-motion intensities. At present time, however, OpenQuake does not have the capability of explicitly incorporating correlations in the capacities of different structures or explicitly considering the time dependence of exposure, structural vulnerability, and seismic hazard.

More recently, Heresi and Miranda¹⁸ introduced an extension of the PEER PBEE framework for application in groups of structures within a region, termed RPBE. The RPBE approach incorporates: (1) spatial correlation of ground-shaking intensities during an earthquake; (2) building classification systems and detailed models of structural archetypes; (3) spatial correlation of damage and losses at different structures; and (4) innovative regional decision variables to quantify the seismic risk of a group of structures on a regional scale. These regional risk metrics include total economic losses, the total number of fatalities, total downtime, the total number of damaged/nonfunctional structures, the total number of displaced families needing emergency shelter, among others. Recognizing the computational complexity arising from several integrals in computing regional seismic risk metrics, the authors suggest employing Monte Carlo simulations for their evaluation. While the RPBE approach currently represents a stationary assessment of seismic risk, it serves as a foundational framework for future enhancements, enabling the incorporation of the time dependence of exposure, structural vulnerability, and seismic hazard into regional seismic risk metrics.

2.2. Assessment of temporal evolution of seismic hazard

Seismic hazard analysis (SHA) is used to estimate intensity measures of strong motion (e.g., PGA, PGV) at a specific location. A SHA is typically used for decision making for social or industrial purposes. It has a variety of applications, such as assisting in structural design guidelines or determining reinsurance premiums and has significant scientific value. It is a multidisciplinary science that integrates many fields of study, such as geology, tectonics, seismology, geodesy, statistics and engineering seismology.

There are two approaches to determine it: deterministic seismic hazard analysis (DSHA) and probabilistic seismic hazard analysis (PSHA). The former estimates the strong motion parameters for the maximum credible earthquake at the closest possible point without considering the probability of its occurrence during a specific exposure period. In contrast, the second approach integrates all potential earthquakes at different locations during the exposure period of interest, considering their associated uncertainties, including epistemic and random uncertainties. The former are inherent in the assumed models of seismicity, zonation, and ground motion prediction, whereas the latter are associated with errors or deviations of the actual data around a given model.

PSHA calculations have undergone numerous advances since Esteva⁴⁹ and Cornell⁵⁰, combining multiple components and their uncertainties. In general, they comprise three elements: (1) seismic source characterization; (2) ground motion characterization; and (3) uncertainties. Cornell⁵⁰ introduced seismogenic zone considering a homogeneous seismicity within a given perimeter based on geological and geophysical information. However, controversies arose due to the establishment of arbitrary boundaries, which led to the adoption of smoothed seismicity, eliminating the need for such boundaries^{e.g., 66,67}. In addition, seismic hazard calculations have been incorporated into fault-sources based on geologic and paleoseismologic data^{e.g., 68,69}. This approach has evolved from segmental and repetitive fault analysis to a multi-segment fault approach, as observed in New Zealand in 2016⁷⁰.

Traditionally, seismic hazard estimations have adopted a time-independent approach, assuming that earthquakes follow a Poisson distribution, which required the elimination of aftershocks and foreshocks using spatiotemporal windows such as those of Gardener and Knopoff⁷¹ and Reasenber et al.⁷². However, Bufe et al.⁷³ argued that time-dependent models outperform time-independent models. The first descriptions of time-dependent recurrence in temporal models were those of Rikitake⁷⁴ who used a Gaussian distribution, and Hagiwara⁷⁵ who used lognormal, gamma, and Weibull distributions. Subsequently, Matthews⁷⁶ proposed the Brownian Passage Time (BPT) model, and Zöller⁷⁷ demonstrated the suitability of BPT models to relate paleoseismicity to the instrumental record of different magnitudes. Acuña et al.²⁶ showed how the hazard changes as a function of time since the last earthquake, considering positions 20 years before and 60 years after the 1960 Valdivia, Chile earthquake.

Another leap improvement for PSHA calculations was the deviation from the ergodic assumption^{e.g., 78–81}. These partially ergodic proposals rely on the particularities of the response of a given site (site-to-site reduced variability) or those of a given source-to-site path (path-to-path reduced variability) which need ground motion models capable of separating the uncertainties related to each of these effects^{e.g., 82,83}. For the case of the site component, there are some alternatives which mostly involves 1D nonlinear site response analyses^{e.g., 84–86}, while the path-to-path effect is generally approached empirically by developing ground motion models with a domain divided into cells which describe systematic path effects⁸⁷.

Liquefaction is a major cause of damage, and a great hazard for civil infrastructure. It cannot be overlooked by EASER due to the relevance for the decision-making process and the widespread impact on society. The seismic demand has two main roles in a liquefaction assessment: (1) estimate what is the threshold intensity measure, not necessarily PGA, that can cause liquefaction or surface manifestation of it at a given site or layer, and (2) estimate the likelihood of this demand occurring at this given site. The first question can be subdivided into two questions: (1) what intensity measures best explain the triggering of liquefaction in subduction environments? a question that has been readily answered for shallow crustal environments^{88,89}, but it's still open for subduction zones; and (2) what is the intensity level that can cause relevant liquefaction and settlements? Several authors have addressed this for shallow earthquakes^{88,90–92}, yet the evidence shows that current methods are inaccurate by more than 40%^{37,38}. While for CPT-

based methods, there is no published research that is specific to subduction zones, preliminary analyses with the available dataset³⁷ show similar results. Finally, non-ergodic or partially-ergodic models for liquefaction triggering in subduction zones are non-existent. Despite the progress made in seismic hazard estimation, all the above advances result in a stationary calculation of seismic hazard. None of them incorporate variations from current conditions or anticipate changes that will occur in the future, highlighting the importance of seeking calculation methods that provide a time evolution of seismic hazard. This includes accommodating advances in the understanding of seismic phenomena over time.

2.3. Assessment of temporal evolution of exposure

Over time, exposure evolution is increasingly evident in large cities. Hence, exposure must be considered a dynamic process due to rapid urbanization, depopulation of rural areas, and all the changes associated with the evolution of human settlements⁹³. The challenge is, therefore, to find innovative methods to automate remote sensing processes and growth forecasting models, based on the reconstruction of historical data and future growth models. For instance, integrating different data sources, statistics, population, cartographic files, satellite imagery, and urban planning can lead to a more exhaustive assessment of urban growth estimates. Indeed, the analysis cannot only be done to the past, but also the simulation of future urban built-up areas can be done using models such as the proposed in Mesta et al.⁹⁴.

As mentioned, one of the tools that can be used for monitoring the urban exposure over time are optical satellite sensors such as Landsat or Sentinel-2. The arrival of Sentinel-2 has marked a milestone in urban settlement monitoring, increasing the number of spectral bands and higher spatial resolution (i.e., 10-20 m) with respect to Landsat data, along with a review time of 5-days since March 2018 for the entire globe (including the Chilean territory). As a consequence of the availability of new data, new artificial intelligence (AI) approaches that exploit multi-temporal imagery have been developed and implemented to delineate the current extent of urban settlement. These models are performed by exploiting a series of reference building schemes to identify a suitable and robust subset, ultimately resulting in the built environment's temporal evolution e.g., 95–98.

In line with the availability of new data sources, image processing methods have improved in recent years based on the joint exploitation of big data, information technology, and AI⁹⁹. In many studies, deep learning is useful for land cover classification¹⁰⁰, with successful applications in urban areas^{101,102}, agricultural land¹⁰³, and wetlands^{104,105}. The deep learning models used for land cover classification are based on artificial neural network algorithms¹⁰⁶, including convolutional neural networks (CNN)^{107,108}, recurrent neural networks (RNN)¹⁰⁹, and generative adversarial (GAN)¹¹⁰. CNNs are currently the most used in remote sensing image processing^{101,111,112}, and their structure consists of an input layer, several convolutional and pooling layers, and a final output that is used for the classification of the image¹¹³.

Some Earth Observation (EO) platforms offer integrated global datasets, thematic products, and services that leverage observation capabilities and big data analytics to process and analyze information⁹⁹. This allows users to acquire valuable information on the dynamics and trends of urban exposure, making it a valuable resource for seismic risk mitigation. These platforms represent an open virtual environment that allows the use of dedicated functionalities and services for access, processing, analysis, and visualization of urban data, such as the evolution over time of exposure (Figure 3). Elements of this scope are addressed through related activities

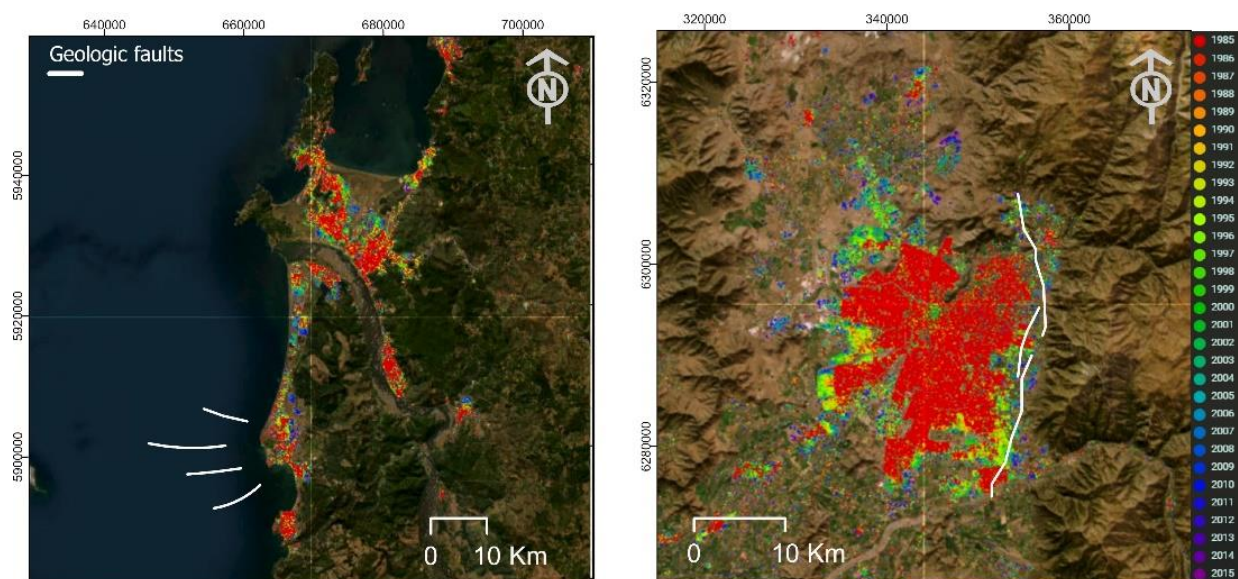


Figure 3: Evolution of exposure over time between 1985 –2015 for Metropolitan area of Concepción, and Santiago, Chile, using historical satellite data from Landsat. Colors represent different epochs.



and platforms, such as Google Earth Engine¹¹⁴, Copernicus Data and Information Access Services – DIAS¹¹⁵, Copernicus Germany Data and Exploitation Platform – CODE-DE¹¹⁶, where all Sentinel-2 scenes are available in a high-performance rendering environment.

Mobile phone data is another data source that can be used to model the evolution of exposure. This information can be used to extracting information from the density profiles of people in a given area at various times, which can even be employed to track human daily commutes by analyzing the spatio-temporal patterns and trajectories of mobile subscribers^{117–119}. Although this type of databases are not publicly available, detailed data from mobile phone companies is available to the research team through a non-disclosure agreement (NDA) with a large private operator in Chile.

Because the use of imagery and advanced processing tools (e.g., AI algorithms) is relatively recent, their specific applications are still being explored. In this regard, they can be extensively used to gather comprehensive information from large urban settlements to develop time-dependent exposure models and tools for forecasting the analyzed areas' exposure evolution. Furthermore, better results can be achieved if mobile phone data is included in the process, which can result in city-specific exposure models and technological tools to update them (or even self-updating tools). These models will be pivotal to a realistic computation of earthquake risk that accounts for the many variables that influence the amount of people and structures that would be expected to be subjected to an earthquake scenario in the future.

2.4. Assessment of temporal evolution of structural vulnerability

Structures evolve due to different factors, such as structural modifications, sequences of earthquake ground motions, aging of materials, and attack of biological and chemical agents. For this reason, a time-dependent vulnerability assessment is necessary due to the dynamic nature of the factors involved. In reinforced concrete and masonry structures, earthquake and corrosion are primary factors that damage them^{120,121}. In this context, different authors have studied the evolution of structural vulnerability through time, which is conceptually represented in Figure 4(a) for the case of sequences of earthquake ground motions. In a seminal work, Mahin¹²² performed nonlinear dynamic analysis of single degree of freedom models subjected to sequences of synthetic ground motions, concluding that the ground motion duration might significantly affect the measured inelastic displacement and ductility demand, particularly on stiff structures. This observation is particularly relevant within the Andean seismogenic context for subduction zones, mechanism responsible for the longest earthquakes recorded by modern sensing technologies. Later, Luco et al.¹²³ employed nonlinear SDOF models subjected to sequences of mainshocks and aftershocks to assess the residual capacity of structural systems after an earthquake. Employing a steel moment-resisting frame as a case study, the authors highlighted that aftershocks also significantly degraded the system's lateral strength and stiffness. Later, Iervolino et al.¹²⁴ investigated vulnerability evolution driven by earthquake ground motion sequences using a Markov chain approach and stochastic models. Trevellopoulos and Guéguen²³ also used the Markov chain method to investigate the variation of structural vulnerability with a focus on evaluating post-earthquake safety. The authors found that the elongation of the fundamental period was a suitable indicator of the damage in a building, and that can be used to probabilistically assess the effects of earthquakes. Lallemand et al.²² studied the effect of incremental building expansions (gradual addition of new rooms or structures to existing buildings, often without strict regulations) into fragility with large-scale risk evaluations using a Markovian framework. The approach was used with a hypothetical case study in Kathmandu, Nepal, to conclude that the incremental building process can significantly increase the expected number of collapsed buildings within a community with time. The authors remark that this framework can also be used to evaluate retrofit's influence on vulnerability. Recently, several studies have used similar approaches to investigate further^{e.g., 125} showing, for instance, that the collapse probability in 100 years of a building increase by 63% if mainshock-aftershock sequence is considered instead of solely mainshock. Likewise, only recently, the effects of earthquake duration on structural vulnerability have been quantified to reveal up to 40% reduction on the collapse capacity of a structural system¹²⁶. These earthquake duration effects are expected to play a role in assessing the evolution of structural vulnerability, particularly on buildings subjected to several large earthquakes (e.g., subduction megathrust events).

Additionally, structural reinforcement corrosion produces significant damage and maintenance costs^{120,121}. When speaking of reinforced concrete and masonry structures in populated cities, the corrosion progression can be linked to the carbonatation of the cement paste (i.e., penetration of CO₂), coastal settlements, and chloride ions intrusion¹²⁷. The loss of resistant steel section due to corrosion reduces the strength and ductility of structural members^{128–130}, resulting in an increment of vulnerability. As a matter of fact, the intrusion of chloride ions and CO₂ varies on time depending on the concentration of pollutants in the environment nearby the affected structures^{120,127,131,132}, implying that the vulnerability change also evolves in time. Furthermore, the corrosion advance benefits from the occurrence of earthquakes, since seism-induced cracks become new ways for the intrusion of the compounds involved in the steel corrosion reaction (i.e., water, chloride ions, and CO₂). One way to intend preventing corrosion issues is to adopt a prescriptive design, although more robust designs can be achieved using probabilistic methodologies, such as that proposed in the DuraCrete project¹³³ and later adopted in the Model Code for Service Life design¹³⁴. This probabilistic approach has been used in different studies to assess its relationship with structural vulnerability evolution. For instance, Ferreira et al.¹³⁵ used Monte Carlo simulations to model the corrosion on a concrete harbor structure in Norway. Some reproduced scenarios raised concern, since the collapse probability reached 10% after 10 years of service, raising the need for the existence of a maintenance program. Ghosh and Padgett¹³⁶ studied the life-cycle effects of corrosion on an RC bridge's pier and obtained a significant increase in the whole bridge system vulnerability over time due to ageing. For example, after 75 years of exposure to chlorides the median PGA value for the

collapse limit state decreases by 27%. In other words, a significantly lower ground motion intensity is needed to reach the median probability of collapse with respect to the pristine structural system, as illustrated in Figure 4(b) for an extensive damage state. De Domenico et al.¹³⁷ investigated the vulnerability of bridges exposed chlorides by performing NLTH analyses considering the losses of rebar sections experimentally measured in the Zapulla multi-span viaduct (Italy), a bridge of about 50 years old. The authors concluded that the corrosion generated during the service life produced a reduction of the seismic safety margin of about 40%. Bossio et al.¹²⁸ studied the evolution of vulnerability of a two-story regular moment-resistant frame reinforced concrete building in Italy of different ages (associated to concrete quality) and under different rates of carbonation. The authors performed non-linear push-over analyses to evaluate the degradation of the lateral capacity of the study cases, obtaining that the age of the structure and the environmental pollution are determinant for the structural safety. For instance, the case study built before 1972 was unsafe for earthquakes with a return period of 475 years regardless of the CO₂ penetration rate, but this only occurred in the worst pollution case in the structure built after 2008.

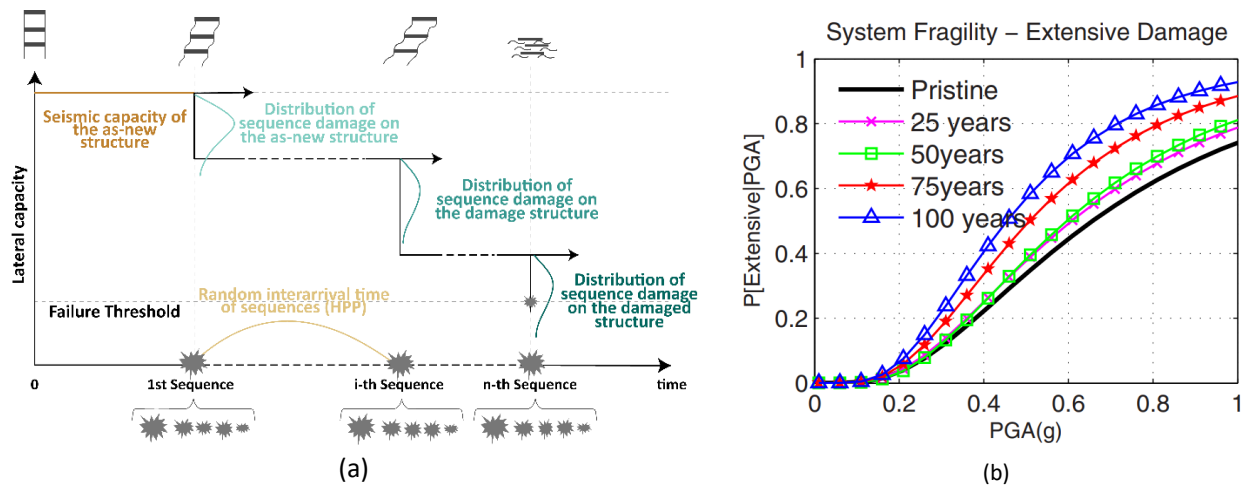


Figure 4. (a) Deterioration process for a structure subject to cumulative earthquake damage. Adapted from Iervolino et al.¹²⁵; (b) Example of time-dependent seismic fragility curve¹³⁶.

In the process of evaluating the evolution of vulnerability, suitable structural models are required. In this context, different authors have studied the effects of employing models with different levels of detail in representing the behavior of structures. For example, Single Degree of Freedom (SDOFs) models have been used to assess the effect of earthquake sequences on the seismic response of buildings^{122,138–142}, concluding, in general, that successive events lead to higher structural damage than single events. Other studies^{123,143,144} have used more complex Multi-degree of Freedom (MDOF) models with similar purposes, generally lumping elements behavior via macro-elements representing complete stories. From these studies, it was also observed that earthquake sequences generated higher damage than isolated events. Furthermore, it was reported that single events with intensity equivalent to a sequence of earthquakes did not produce the collapse of structures, but the sequences of events did induce the failure. On the other hand, even more detailed models have been implemented of RC structures^{145, e.g., 146–148}, masonry structures^{e.g., 149–152}, and steel structures^{e.g., 153} to evaluate earthquake cumulative damage. These more detailed models have facilitated the assessment of damage exhibited by different structural members and also helped to generate design improvements and better retrofit strategies^{e.g., 154}. Thus, more complex models provide more detailed information at a higher implementation time and computational cost. The validity of results obtained from models is usually conducted by comparing them with experimental information. A common approach to validate models at an element level are destructive testing of specimens, such as reinforced concrete walls (undamaged walls¹⁵⁵; walls with previous damage^{156,157} and masonry walls^{158–160}). However, this approach is expensive, time-consuming, and constrained by the size limitations of testing facilities. In this context, non-destructive testing methods are attractive for characterizing existing buildings since they are easier to carry out and less expensive. One of the most common non-destructive techniques used to determine existent building dynamic properties, such as fundamental periods and associated damping ratios, is the Operational Modal Analysis (OMA)¹⁶¹. In this method, several synchronized accelerometers are positioned across the structure and used to measure the micro movements that buildings experience due to ambient vibrations. This information can then be analyzed with techniques, such as auto-regressive models (ARMA) or Short-Time Seismic Frequency Domain Decomposition (STSFDD), to identify predominant vibration frequencies and damping ratios. These methods have been used to characterize reinforced concrete structures^{e.g., 162–164}, masonry structures^{e.g., 165}, and other structural systems^{e.g., 166–168}. Given that tests like these do not generate damage to buildings, the same structure can be tested at different ages, which can inform how dynamic properties evolve over time. Therefore, these results can inform the calibration of numerical models, reducing epistemic uncertainties and increasing accuracy on the actual updated structural behavior of a building.



In summary, adopting a multi-perspective approach for structural representation and characterization will allow to get a comprehensive and reliable model of the change in time of the vulnerability of structures driven by natural hazards like earthquakes and aggressive environmental conditions leading to corrosion.

3. Objectives

Our goal is to assemble a cohesive, multidisciplinary group of researchers to advance the knowledge of the time dependency of seismic risk and to promote an environment that fosters collaboration and innovation, providing an international academic environment to train young earthquake scientists at the highest level. We will integrate seismology, geotechnical, and structural engineering to quantify the interrelation between seismic hazard time-dependency, seismic vulnerability time-dependency, and the exposure evolution over time, with epistemic and aleatory uncertainty considered in every step. Our results will ultimately allow the development of advanced computational tools to aid both public and private stakeholders in assessing more realistically the seismic risk in residential structures and communities within the Andean seismotectonic environment and making informed decisions to mitigate these risks. Although EASER will be particularly interested in the number of displaced families due to damaged residential buildings as the risk metric, the methodology would be extensible to other risk metrics (e.g., total economic losses). Outreach activities will be conducted to raise awareness among society regarding the critical importance of effective seismic risk management, contributing to sustainable development amidst natural challenges.

We will focus on four specific objectives (SO):

- SO1:** Develop time-dependent models of seismic hazard within the Andean subduction context, considering the time between earthquakes, the temporal evolution of locking degrees at the source, of path characteristics, and of site response effects.
- SO2:** Evaluate the evolution of structural vulnerability over time due to degradation of materials and occurrence of earthquake sequences, both from analytical and experimental approaches.
- SO3:** Assess the temporal evolution of exposure through analysis of publicly available databases and image recognition techniques applied to multispectral satellite imagery.
- SO4:** Develop a set of computational tools to evaluate the effect of time-dependent seismic hazard, structural vulnerability and exposure on seismic risk, considering and propagating their uncertainties. We will use three cities within Chile as case studies for our tool set: Santiago, Viña del Mar, and Concepción.

Following these objectives, we expect our project to result in:

- A novel approach to evaluate hazard, vulnerability, and exposure, which will map into a risk assessment that considers aleatory and epistemic uncertainty within a set of parameters which evolve over time but are currently considered stationary.
- Understanding of the impact of time-dependency on each component of seismic risk.
- Computational tool set that stakeholders can leverage to make informed decisions on risk mitigation strategies.
- An international academic environment to train young earthquake scientists at the highest level, exposing them to different backgrounds, technical approaches, and collaborative research.
- A catalyst for collaboration initiatives with national and international institutions for establishing future strategies for seismic risk mitigation.

4. Hypothesis

This project is underpinned by both a scientific and a technological hypothesis.

The scientific hypothesis is that the seismic risk of structures and communities rapidly changes over time, strongly affected by changes in seismic hazard, seismic vulnerability, and exposure (e.g., population growth). Therefore, seismic risk needs to be studied considering its time-dependency. The main objective of our project is to investigate the specific impacts of each of these variables on the time evolution of seismic risk.

Concurrently, the technological hypothesis of this project is that both public and private stakeholders should consider the temporal evolution of seismic risk in order to make informed and rational decisions regarding risk mitigation strategies for future resilient communities.

5. Methodology

As shown in Figure 5, the workload of this project will be divided into four main Research Lines (RLs), which are designed to address each of the four SOs:

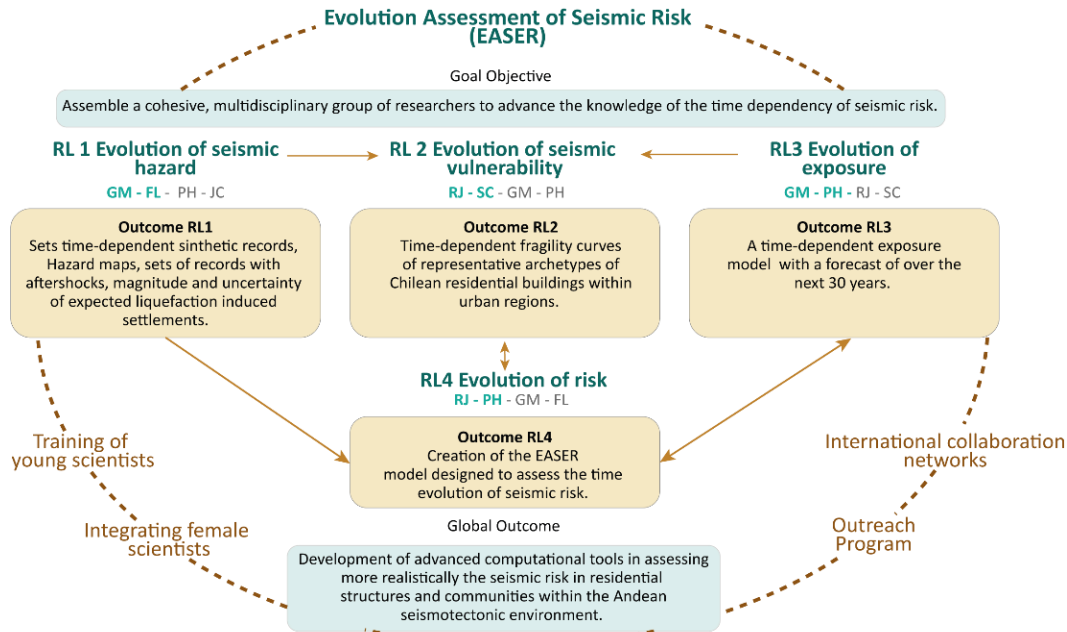


Figure 5. Workflow chart of EASER, showing the relationships between RL's, their main outcomes, and the global goal.

Research Line 1: Time evolution of seismic hazard

This research line has three primary objectives: first, to analyze and characterize the impact of time-dependent recurrence and site effects evolution in the final computed hazard for a given site during a fixed time frame; secondly, to incorporate the evolution of knowledge into the computation of hazard, this is how new faults, or new knowledge about a source, path or site would affect future hazard; finally, the third objective is to produce liquefaction-induced settlement probabilities for a simply parameterized site within a subduction environment. These objectives will turn into relevant products for the computation of time-dependent risk, inputs for RL2, including synthetic records and permanent settlement estimations.

TASK 1.1. Evaluation of Non-Poissonian and Poissonian processes

Seismic hazard analysis will be approached from different points of view. First, in activity 1.1.1, estimates will be made following classical methodologies with time-independent approaches (Poissonian) as is done in most countries of the world, identifying, and characterizing seismogenic sources, paths and site effects¹⁶⁹. This will serve as a benchmark for the following studies.

Activity 1.1.2 will focus on exhaustive surveys of historical and geological data related to past large earthquakes in the Andean subduction zone will be carried out. With this, the frequency, magnitude, and location of past earthquakes will be thoroughly studied (activity 1.1.3). Preliminary results can be found in our previous work by Acuña et al.²⁶, who used the Valdivia Earthquake as a case study. In this topic we will collaborate with Jasper Morneaut, whose team is working on obtaining paleoseismic data through the analysis of turbidites in southern Chile lakes.

Precise relocation of earthquakes plays a crucial role in improving our understanding of locations of fault systems, and more importantly, tectonically stressed regions. The improvement in earthquake location can be achieved by comparing the relative arrival times of seismic waves recorded at multiple stations. This technique has proven to be particularly valuable in megathrusts like the Chilean subduction zone, with notable contributions in Central and Northern Chile¹⁷⁰⁻¹⁷³, providing valuable insights into the seismic behavior and fault interactions within this region. From these results, spatial and temporal variations of parameters such as b-value in different specific regions will be studied. We will analyze different effects that may affect this parameter, such as the removal of aftershocks and foreshocks versus not doing so, occurrence of well-documented seismic crises, such as Maule 2010 (Mw 8.8), Iquique 2014 (Mw 8.2), Illapel 2015 (Mw 8.4), Melinka 2016 (Mw 7.6), and Valparaíso 2017 (Mw 6.9) earthquakes (activity 1.1.4).

In activity 1.1.5, time-dependent analyses will be performed using statistical distributions such as the Brownian Passage Time (BPT), Gamma, Weibull, Lognormal, and ETAS, calibrating the necessary parameters for different sites in the country, considering the paleoseismicity data available. This information will be the input in the development of a time-independent analysis of seismic hazard (activity 1.1.6).



Finally, computational models will be developed to simulate earthquakes and their effects on different site classes to predict the temporal evolution of seismic hazard based on historical and recorded earthquakes, with which we will develop time dependent recurrence models for the entire country (activity 1.1.7).

TASK 1.2. Evolution of site characteristics

With appropriate earthquake rupture scenarios consistent with surface deformation data, and seismicity, we can characterize spatial earthquake hazard; however, other aspects of the hazard are also spatially dependent. Such is the case of wave propagation as well as site effects phenomena. The approach to include this within a spatially consistent seismic hazard is usually to infer attenuation, average stress drops, and site effects along the region of interest. There are two main methods to estimate source, path, and site parameters in a generalized framework: i) The parametric approach¹⁷⁴, which relies on the presumption of the spectral Fourier Amplitude Spectral (FAS) shapes of attenuation with distance, the seismic source and site effects, such as the high-frequency shape given by the spectral falloff ($exp(-pkf)$) given by the k parameter; ii) The non-parametric approach^{175–177} assumes that the attenuation is a smoothly varying function. Consequently, the acceleration spectral amplitudes at frequency f , $U_{ij}(f, r_{ij})$ can be written as $\log(U_{ij}(f, r_{ij})) = \log(A_{ij}(f, r_{ij})) + \log(S_j(f)) + \log(Z_i(f))$, where r_{ij} is the distance at which ground motion was recorded at a station “i” due to a source “j”, $A_{ij}(f, r_{ij})$ is a smoothly varying attenuation function, and $S_j(f)$ is the source scaling function, which depends on the magnitude and stress drop of the earthquake, and $Z_i(f)$ is the site amplification function at station “i”. Depending on the number of recording stations and earthquakes, we can use this relationship to define a set of linear equations, which can then be solved for each A_i , Z_i and S_j , and for each frequency. These sets of linear equations are solved with the singular value decomposition (SVD) method or put forth in a Bayesian inversion method scheme¹⁷⁸. The parametric method has been used extensively in several tectonic regions, however, not in Chile, (apart from Fernández et al.¹⁷⁹, who applied the method to the region of Illapel, Chile), and can easily be extended to the Chile megathrust. This method has the potential to evaluate stress drop variation both spatially and temporally; to be developed in activity 1.2.1. Furthermore, this methodology will be applied considering different time windows in the seismic cycle in Chile, enabling us the quantification of time changes (activity 1.2.2).

Moreover, site characteristics in cities change due to the Consolidation of sediments^{35,180}, which can be caused by human settlements or an increase in the load on the soil’s surface, a decrease in groundwater table which increases effective stresses of the soil and can be augmented by climate change. These changes in site characteristics can be proxied by the change in natural site periods¹⁸¹. Using available data from the last decade in our three case studies cities, we will assess the changes over time in Concepción, Viña del Mar, and Santiago and project these changes over time to account how a site will amplify/deamplify the seismic waves in future events. This task will involve measuring site periods as well as using RL3 results from InSAR measurements to study ground surface deformations over the last decade (activity 1.2.3). In activity 1.2.4., we will model these changes to extrapolate them to other cities. Using the existing site response observatory in the City of Concepción^{182,183}, we will empirically differentiate the evolution of site response in our context from those well-instrumented as well (e.g. EuroSeisTest site; Kik-net).

TASK 1.3. Evolution of knowledge

The design value of the seismic intensity on the PSHA methodology is defined not only by the median of the prediction but also by the uncertainty. In this line, uncertainty can be partitioned into aleatory and epistemic uncertainty, both these uncertainties are included in PSHA. The first one is the inherent randomness in the seismic phenomenon, so, it does not decrease over time and theoretically cannot be reduced, even though, given a large and enough time, all the values of the variables will eventually be sampled. However, on the other hand, epistemic uncertainty can be addressed through continuous improvement in data collection, analysis techniques, and understanding of seismic processes¹⁸⁴. Epistemic uncertainty is associated with the lack of knowledge and can be reduced with further research, advancements in seismic hazard analysis methods¹⁸⁵, or for additional (or new) data (e.g., when the SRF was found to be an active shallow crustal fault in the western region of Santiago, Chile²⁸). In this sense, the boundaries of the knowledge and, in consequence, epistemic uncertainty is time dependent.

Standard deviations are time and space-dependent and could be reduced by discounting the ergodic assumptions. However, the assumption of ergodicity, which suggests that the statistical properties of the earthquake process are time-invariant over long periods, may not hold for all seismic hazard analyses. Therefore, it is necessary to consider temporal and spatial variations. Evaluation of available seismic data on the Chilean subduction zone shows a clear region-to-region variability^{186,187} that must be included in seismic hazard analysis, also new earthquakes reveal new data that can be assimilated into the analysis and lead to updates in the seismic hazard assessment. Especially, megathrust earthquakes that control and have the potential to significantly impact risk hazard assessments, as they can introduce new data points and insights into the understanding of earthquake processes. Moreover, a dynamic update of the weights of the logic tree can be useful due to the amount of data and knowledge accumulated over time.

Considering the limitations of the time-independent Poisson process approximation and the challenges it poses to seismic hazard estimates, there is a need for incorporating time-dependent uncertainty in the PSHA scheme and their validation^{e.g., 188–190}. Epistemic uncertainty could be accounted for by new experimental (or observations) data. Subduction megathrusts are break-points of observational data on knowledge and can provide valuable insights into the temporal variation (and uncertainty) of seismic hazards because they probably control the design earthquake-scenario. Maule (2010), Iquique (2014), and Illapel (2015) earthquakes have all highlighted the dynamic nature of seismic hazard in the subduction zone and could be used to test epistemic considerations before



and posterior to the seismic event, allowing us to examine the temporal variations and update our seismic hazard assessments. Also, new data assimilation must be conducted within a framework that allows the identification of new seismic faults, their geometry, segments, and the assessment of their impact into seismic hazard analysis. New paleoseismic data^{e.g., 26,191} can define a new recurrence of the seismic cycle¹⁹² and cause significant differences in the magnitude-frequency distributions for seismic hazard analysis.

This task uses cumulative data from the previous task and considers (activity 1.3.1) the study and account the uncertainty on each seismic hazard analysis component (i.e., recurrence of the seismic cycle, path, and site conditions); (activity 1.3.2) consider and propagating appropriately these uncertainties onto probabilistic seismic hazard analysis; (activity 1.3.3) estimating the spatial and temporal evolution of the variation of strong ground motion intensities in the Chilean subduction zone to conduct partially- or non-ergodic analyses (site-to-site, path-to-path or region-to-region variability); (activity 1.3.4) the use of case histories (i.e., Maule, Iquique, and Illapel events) to evaluate the evolution of epistemic uncertainty and their influence on seismic hazard estimates, and (activity 1.3.5) generate a model to estimate a new epistemic uncertainty (space and time-dependent) and their impact on updating the seismic hazard for Concepción, Viña del Mar, and Santiago. This task provides inputs for Task 1.4 and adjusted intensities for RL2.

TASK 1.4. Earthquake-induced liquefaction

Here we will evaluate the effects of controlling parameters: Variety of Intensity Measures, CPT Tip Resistance, Shear-Wave velocity, Small Strain Damping, State Parameter assessment in Subduction Environments. A curated database of all field and laboratory data is already available³⁷, and other sources (such as NGL project) will be added, to be able to evaluate the relevance of each controlling parameter. This database is publicly shared to be used by our research team and the entire engineering community. An advanced CPT equipment from the main university is available for this task and technical aid is considered for part of the field work. International collaborator Dr. Rodriguez-Marek will help supervise this task.

Three-dimensional imaging of shear-wave velocity (Vs) and small strain damping will be performed at selected locations, along with profusive CPT testing to validate and compare assessment methodologies. These locations will consider sites with (yes), and without (no) observed settlements, these will be the Los Presidentes Condominium (yes), Valle Noble Alicante (yes), Hogar Freire (no), Valle Noble Madrid (no), Santo Domingo (yes), Chiguayante (no), Villa Hermosa Hualpén (yes). This task will follow state-of-the-art methodologies^{193–195} to estimate Vs and damping at different scales. These field measurements will be performed with 7 to 9 independent Tromino seismometers spaced at different distances to capture different wavelengths. This imaging exercise will be assisted by the methods and results of project Fondef ID2210032 where both RL1 lead researchers participate.

Despite the availability of complex numerical models and sophisticated computational architecture, reliably reproducing the different liquefaction phenomena simultaneously is challenging, even under simple field conditions. Therefore, we will prioritize behaviors and make simplifications in modeling to evaluate the relative merit of each of the controlling parameters mentioned above. Since diffracted and scattered elastic waves in irregular media alter soil movement¹⁹⁶, the spatial variability measured in-situ may have a direct impact on liquefaction susceptibility and deformations¹⁹⁷, and the amount of these impacts will be validated with the site data and analyzed parametrically as well. Within this task we will (activity 1.4.1) compute state parameter from CPT at the 7 locations in Concepción and evaluate the spatial dependence (activity 1.4.2) Compute shear-wave velocity 3D field at 4 of these locations and evaluate time evolution of these by using ORSCA (the site response observatory at the Concepción basin) and comparing with published data by the research team¹⁸¹. (activity 1.4.3) Generate a model, and a computational tool, to estimate the magnitude and the uncertainty of expected liquefaction induced settlement given different kinds of data and considering time variation throughout.

Research Line 2: Time evolution of seismic vulnerability

This research line has two primary objectives: first to investigate the temporal evolution of structural vulnerability and characterize it through analytical time-dependent fragility functions; and second, to develop guidelines for assessing the vulnerability of existing structures considering time-dependent effects. Our primary focus will be on residential buildings made of reinforced concrete and masonry, as they are the most prevalent materials in residential housing in Chile, particularly in the testbed cities¹⁹⁸. Employing a comprehensive approach, we will integrate both analytical and experimental methods to ensure robust and reliable results, serving as input information for subsequent steps of the risk assessment process. The methodology is structured into four main tasks: (i) archetype buildings characterization, (ii) development of detailed analytical models including time-dependent effects, (iii) experimental characterization of time-dependent effects, and (iv) development of time-dependent fragility functions.

Task 2.1. Archetype buildings characterization

Based on the exposure model by Santa-María et al.¹⁹⁸, our focus will be on a simplified subset of six building typologies, considering different construction materials and number of stories. Masonry structures will be classified into two typologies: masonry houses (up to two stories), and masonry apartment buildings with three to five stories. RC structures will be classified into four different typologies: RC houses (up to 3-stories high), low-rise RC apartment buildings (3–9-stories high), mid-rise RC apartment buildings (10–24-stories high), and high-rise RC apartment buildings (taller than 24-stories). Additionally, we will categorize RC structures into two targeted-ductility levels to address variations in seismic provisions. These considerations arise from design modifications introduced in Chile after the 2010 Maule Earthquake for this type of structures (decrees DS 60 and DS 61). Thus, the first activities correspond



to the characterization of each building typology (2.1.1 for masonry and 2.1.2 for RC) in terms of structural properties such as structural system, floor plan area, slenderness ratio, wall density ratio, among others.

In this study, we will select a set of existing buildings with structural characteristics that are representative of the respective building typology, which will be called archetype buildings. The selection of existing buildings is particularly valuable, considering that some of them may have experienced different and several past earthquakes due to Chilean seismicity. VMB Engineering, a private consultancy with extensive experience in structural analysis and design of residential buildings in Chile, which is also an Interested Institution in this project, will contribute to the selection of the typical RC buildings. They will also provide detailed information about selected RC buildings (i.e., drawings, calculation sheets and analytical models). In the case of RC houses and masonry systems, the research team will gather detailed information from public records available in different municipalities or local authorities and scientific literature. Consequently, the activities associated with the selection of archetypes are 2.1.3 for masonry and 2.1.4 for RC.

Task 2.2. Development of detailed analytical models, including time-dependent effects

Different factors influence the evolution of buildings' structural capacity over time. This project will concentrate on the most significant factors within the Chilean context, which are the effects of cumulative earthquake damage, and chloride- and carbonation-induced corrosion. These time-dependent effects will be included in the analytical models of the archetype structures. Detailed non-linear finite element models (FEM) of each of the archetype buildings will be developed, corresponding to activities 2.2.1 for masonry and 2.2.2 for RC archetypes. The models will consider inelastic behavior at a material level through a specific non-linear constitutive model for each material, as done in previous investigations by the research team^{e.g., 146,199}. Each building model will first consider the nominal material properties, and then the time-dependent effects will be included.

The effect of previous earthquake damage will be evaluated by conducting Non-Linear Time-History (NLTH) analyses of the selected building archetypes under various scenarios by combining sequences of seismic records. A set of sequential records will be defined based on results from RL1 (activity 2.2.3), to evaluate the reduction in stiffness, strength and/or deformation capacity due to previous earthquake damage in time (activity 2.2.4). In these analyses, liquefaction-induced settlements will be considered as a random variable taken from RL1 results.

Regarding reinforcement corrosion, two factors will be accounted for: carbonation of cementitious materials (concrete and mortar) and chlorides exposure. In this study, the advance of the carbonation front and chloride concentration in the cementitious matrix over time will be estimated using the guidelines of Model Code 2010¹³¹ and Model Code for Service Life¹³⁴, and suggestions and simplifications proposed in Chen et al.²⁰⁰, Angst et al.²⁰¹, and Geiker et al.²⁰². For masonry, similar time-dependent degradation models will be used^{e.g., 150,158,199}, complemented with considerations stated in Ghiassi & Lourenço²⁰³. Thus, the assessment of the degradation of masonry and RC mechanical properties due to corrosion will be associated to activities 2.2.5 and 2.2.6, respectively. Accordingly, results from corrosion precursor assessment will be used to modify models of building archetypes to properly account for the decrement in mechanical properties.

Finally, different detailed FEM of each archetype building will be developed, considering different material properties to incorporate both time-dependent effects. Non-linear static (pushover) and dynamic analysis will be carried out, which correspond to activities 2.2.7 for masonry archetypes and 2.2.8 for RC archetypes, respectively. The performance-based earthquake engineering (PBEE) approach will be followed, to evaluate the expected performance under various earthquake scenarios and damaged conditions considering time-dependent effects. These analyses will also allow to identify the best repair and/or retrofit strategies to reduce the seismic risk of the buildings, based on observed damage of critical structural elements, which will be executed in activities 2.2.9 and 2.2.10 for masonry and RC archetypes, respectively.

Task 2.3. Experimental characterization of time-dependent degradation:

To calibrate and validate the developed FEM models, experimental information will be gathered. Sets of Tromino seismometers will be used to carry out ambient vibration tests to get information that will be processed through Operational Modal Analyses (OMA) methods for the empirical measurement of dynamic properties, including dominant vibration frequencies and associated damping ratios. This experimental characterization will be conducted on buildings with different ages, including recently constructed ones, those with intermediate ages that have not experienced major earthquakes (e.g., 3-years old buildings), and others that have endured one or more major earthquakes. When possible, the archetype buildings will be monitored. The execution of vibration tests on masonry and RC case studies will be executed in activities 2.3.1 and 2.3.2, respectively, and the analysis of results of the tests with the activities 2.3.3 and 2.3.4, respectively.

The gathered experimental information will serve to: (i) evaluate differences between projected (i.e., estimated at design stage) and as-built dynamic properties (activity 2.3.5), and (ii) assess time-dependent effects studied in this project, namely earthquake-induced cumulative damage and corrosion-induced damage (activity 2.3.6). The trends of the evolution of dynamic properties experimentally measured will be used to adjust modelling assumptions.

Task 2.4. Development of time-dependent fragility functions

Simplified MDOF models of each archetype building will be developed in the software OpenSees²⁰⁴, which will lump the inelastic behavior at a story level (activity 2.4.1 for masonry and 2.4.2 for RC). Each simplified model will be calibrated with the detailed FEM



counterpart, to generate simpler models for the series of NLTH needed to develop fragility functions. These simplified MDOF models will be subjected to NLTH analysis considering a set of seismic records provided by RL 1 to develop fragility analysis (activity 2.4.3). To account for the dynamic properties of the different building typologies, we will consider peak ground acceleration (PGA), spectral acceleration at periods of 0.3 seconds ($S_a(0.3s)$), 0.6 seconds ($S_a(0.6s)$), and 1.0 second ($S_a(1s)$) as intensity measures. A multiple stripe analysis (MSA) approach will be used, where a specified set of IM levels is selected, each of which has a unique ground motion set^{205,206} to perform the structural analysis. We will consider at least twelve evenly distributed IM levels, each with at least 30 seismic records. For each IM level, records closest to the specific IM level will be chosen, aiming for a scaling factor as close to 1 as possible to avoid any bias in the structural response^{207,208}. Each nonlinear model will be subjected to NLTH analysis considering each seismic record, and the most representative EDP of the structure will be calculated, say roof drift, and maximum inter-story drift, among others (activity 2.4.4). To determine the damage state according to each maximum EDP, limit thresholds available for Chilean RC and masonry walls will be considered^{145,199}. The probability of exceeding each damage state given a ground motion intensity measure level will be obtained, and a logarithmic curve will be adjusted to the results, which corresponds to the fragility curve for the building archetype (activity 2.4.5). Finally, we will analyze the temporal evolution of the fragility functions of each building archetype. The time-dependent fragility function of the archetype buildings will be correlated with the time-independent fragility functions of the different building types (activity 2.4.6).

Research Line 3: Time evolution of exposure

This research line will investigate the temporal evolution of urban exposure, focused on residential structures in the Chilean cities of Santiago, Viña del Mar, and Concepción. It will encompass aspects such as the number of residential structures, their locations, and various properties including material of construction, age since construction and building height. This research line is divided into four main tasks, with a special emphasis on the uncertainty associated with exposure, both at the present time and in future forecasts:

TASK 3.1. Historical aerophotogrammetry and satellite image processing

To understand in deep the temporal evolution of exposure, a documentary archive of the main historical sources of aerial photogrammetry available for each of the case study cities will be compiled and catalogued (activity 3.1.1). The data set to be collected will include historical cartography, urban plans, orophoto-mosaics available in Chilean governmental institutions (e.g., MOP, SECTRA or the Geographic Institute Military (IGM)), and topographic databases (e.g., aircraft LiDAR). In addition, we will use advanced satellite image processing techniques, using SAR (synthetic aperture radar) images (such as Sentinel 1) and optical images (such as Landsat or Sentinel 2). Once collected, the satellite image data will be processed (activity 3.1.2) in an open-source toolbox, involving cluster computing and process automation for SAR Interferometry²⁰⁹. Optical images will then be processed through RGB multispectral analysis, generating thematic information on land use and the development of the built environment, based on data collections from multiple sources (Landsat or Sentinel 2). This information will then be used to reconstruct urban settlements (activity 3.1.3), through historical photointerpretation techniques and digitization of data sources, providing new thematic mapping results over time, including the heights of buildings from software with AI plugins available, in QGIS, MAPFLOW, ARCGIS. Epistemic uncertainty will be explicitly incorporated into each of the activities, and propagated to the exposure model.

TASK 3.2. Gathering and analysis of publicly available building data

In parallel to Task 3.1, a residential building exposure model for the case study cities will be developed based on publicly available building data, employing the building typologies proposed by Santa-María et al.¹⁹⁸. This model is developed using freely available data, a condition that is not always met by satellite images. This approach will allow us to extend the exposure model to other study case cities if required, and also to complement the data gathered in task 3.1. The main source of publicly available information will be the property cadaster provided by the Chilean Internal Revenue Service (Servicio de Impuestos Internos, SII), which contains detailed information of each construction unit within a real state property, including address, year of construction, occupancy class, construction material, construction quality, and built area, among others. The information from the SII database is geo-referenced and assigned to a census block according to the definition of the national census data²¹⁰. Information from this database will be gathered (activity 3.2.1) and processed to create the exposure model. This process consists in geolocating each construction unit (activity 3.2.2) and estimating the building height by analyzing the apartment numbers listed in the address of each construction unit within the database (activity 3.2.3). The research team will also use mobile phone data, available through an existing agreement with a major operator in Chile, to validate other data sources and as an additional tool to estimate population growth, location, and even seasonal habits. Uncertainties in these estimations will be explicitly accounted for and propagated to the exposure model. It is worth mentioning that this methodology is very cost-effective, easily scalable, and allows for analyzing the evolution of exposure in different cities based on the historical availability of data (at least 5 years back).

TASK 3.3. Development of enhanced time-dependent exposure model

Combining the results of Tasks 3.1 and 3.2, we will create an enhanced exposure model of the three case study cities (activity 3.3.1). From here, we will develop a forecasting model capable of estimating the future exposure of these cities based on population growth estimates, urban plans, and past history of the cities (activity 3.3.2). This model will classify each building type according to the



number of stories as low-rise (L) for 1 to 3 stories, mid-rise (M) for 4 to 7 stories, or high-rise (H) for 8 or more stories. The temporal analysis of the exposure contained in the generated databases will be complemented with the information available in the Communal Plans (PRC), urban planning instruments which define a series of urban planning regulations, including, among others, permitted and prohibited land uses, areas of risk and protection, building heights, densities, and maximum building capacity. This will allow us to reconstruct and measure changes in exposure over time, for each of the case studies, and also enables the assessment of potential scenarios related to city growth or densification. Moreover, uncertainty in future estimates will be explicitly incorporated in the model.

TASK 3.4. Validation experiment with historical data

In this task, the exposure evolution model developed in Task 3.3 will be subjected to a validation experiment. In this, historical data of buildings until 1985 in the study case cities, will be used to forecast the actual exposure registered. The threshold for splitting the database was selected since it coincides with two major events: (i) the Viña del Mar 1985 earthquake; and (ii) the year since Landsat imagery is available for Chile^{211,212}. The Viña del Mar 1985 earthquake is significant because it generated collapse and damage on several low-rise buildings in two of the case-study cities (Viña del Mar and Santiago)^{213,214}. Thus, the first activity of this task (activity 3.4.1) will be to identify the corresponding information from databases produced on previous tasks.

Subsequently, the information of buildings before 1985 will be introduced into the developed exposure model for the case study cities (result of task 3.3) to predict the current exposure (activity 3.4.2). This will allow us to verify the accuracy of forecasts of our model for different periods up to a time window of almost 40 years (1985 - 2024). Then, we will compare the results of the forecast and the exposure at the present time (obtained in previous phases) (activity 3.4.3). This comparison will allow us to assess the influence on the variation along time of building exposure to natural factors (e.g., earthquakes) or human factors (i.e., urban planning policies). It is also expected to get information about the accuracy of our predictions and the most determinant factors that influence the evolution of the exposure.

Research Line 4: Development of computational tools for evaluating time-dependent seismic risks

The final research line of EASER focuses on the technological aspect of the project, and it is divided into two primary tasks. The first task involves the development of a set of tools for the time-dependent evaluation of seismic risk, while the second task entails using these tools to compare future seismic risk mitigation strategies in the cities of Santiago, Viña del Mar, and Concepción. The utilization of these tools aims to provide stakeholders with a foundational framework for their own time-dependent risk assessments.

TASK 4.1. Development of computational tools for time-dependent seismic risk evaluation

This task integrates the scientific knowledge acquired from the three previous Research Lines (RLs). Initially, all computational tools, including databases, functions, and models, from the prior RLs will be systematically reviewed, standardized, curated, and uploaded to public repositories such as GitHub or DesignSafe (activity 4.1.1). Subsequently, the mathematical formulation of the convolution of exposure, hazard, and vulnerability into risk will be extended to incorporate the time variable (activity 4.1.2). The resulting computational model based on this extended formulation, entitled EASER model, will also be made accessible through public repositories.

TASK 4.2. Test developed tools in case study cities

This task encompasses three case studies: Santiago, Viña del Mar, and Concepción. Santiago, being the capital and most populated city of Chile, will be studied alongside Viña del Mar and Concepción, two major urban regions in the country in terms of both population and economy. Initially, a stationary seismic risk assessment will be conducted in these cities, serving as a benchmark representing the current state-of-the-art (activity 4.2.1). Subsequently, the EASER computational models and tools will be employed to forecast the evolution of seismic risk over the next 30 years, facilitating a comparison against the previous stationary assessment (activity 4.2.2). Finally, various seismic risk mitigation strategies will be evaluated using the EASER models (activity 4.2.3). These strategies include seismic retrofitting programs for the most vulnerable existing buildings, updates to seismic codes to enhance the seismic performance of new buildings, alterations in urban planning policies (e.g., restrictions on building heights in specific areas with soft soil deposits), and regional-scale site consolidation programs aimed at reducing site amplification of ground shaking intensities.

6. Expected Results

The EASER project has expected results of different kinds (i.e., scientific, technological, dissemination, human training, and collaboration):

1. The project's scientific contribution lies in the formal assessment of an integrated methodology that incorporates the time dependency of seismic hazard, vulnerability, and exposure within the Andean seismotectonic environment. This scientific advancement aligns with the technological contribution of the project: the EASER model. This novel model will empower public and private stakeholders to evaluate seismic risk and devise strategies (e.g., site consolidation programs, building



retrofit programs, seismic code updates, urban planning) to attain time-dependent risk targets. The anticipated outcomes from each Research Line (RL) are as follows:

- a. RL1 – Seismic hazard maps that account for time-dependency; a set of mainshock-aftershock ground motion records; magnitude and uncertainty estimates of anticipated liquefaction-induced settlements.
 - b. RL2 – Time-dependent fragility curves of representative archetypes of Chilean residential buildings within the studied urban areas.
 - c. RL3 – A time-dependent exposure model with a forecast over the next 30 years.
 - d. RL4 – Creation of the EASER model designed to assess the time evolution of seismic risk.
2. Scientific productivity: For the entire project, we plan 23 collaborative research articles as detailed in letter j.
 3. Extension and dissemination of results: we envision follow-up collaborative research projects with both public and private stakeholders based on common interests on seismic risk assessment.
 4. Human capital training: the EASER project will fund 2 PhDs students, 6 MSc students and 1 postdoctoral researcher. More researchers with funding outside EASER (e.g., ANID fellowships for PhD and MSc students and ANID Postdoctoral projects) are also going to be trained in earthquake engineering sciences. Equality, diversity and inclusion practices will be considered during the recruitment process.
 5. We will host one in person open workshop per year of the project, a webpage that will be updated throughout, social media content and reels at least 6 times a year, one podcast series on year two, didactic material to convey the relevance of earthquake science, and how to use the available tools (including our developments). In addition we will honour our scientific commitments and present in national and international conferences at least twice a year.
 6. As detailed in letter f, we have a solid national and international network that will be strengthened should EASER be funded. Our goal is to have 18 research stays from students or researchers from our international partners or from our students and researchers to their research centers.

7. Scientific and Technological Merits

The scientific and technological merits of EASER are visually represented in Figure 6, illustrating a hypothetical example of a seismic risk metric's evolution (e.g., expected number of displaced families) in Santiago, Chile. This schematic presentation captures key components of the project's contributions. In this example, the seismic risk increased significantly as the San Ramón Fault (SRF) was confirmed active in 2002, reflecting an evolving seismic hazard (SO1; RL1). Subsequently, the 2010 Maule Earthquake potentially caused damage to various structures, resulting in an increase of seismic vulnerability (SO2; RL2). Furthermore, the example depicts a continual rise in seismic risk over time due to the city's growth, emphasizing the evolving exposure (SO3; RL3). Finally, the example offers a forecast of seismic risk (SO4; RL4), considering two distinct risk mitigation strategies: (1) a retrofit program for older buildings and (2) an update to the seismic code for new constructions. The shaded areas in the illustration represent the aleatory uncertainty associated with each estimation, acknowledging the inherent variability in these risk assessments. Note that both the past and future seismic risk will have epistemic uncertainty (not illustrated in Figure 6). This comprehensive depiction highlights the interconnected scientific advancements and technological applications achieved through the EASER project, providing a holistic understanding of seismic risk dynamics in a specific urban context.

The scientific merit of EASER lies in unraveling the intricate effects of seismic hazard, vulnerability, and exposure evolution on the seismic risk of urban regions. Some of the questions that will be explored are: Which factor—seismic hazard, vulnerability, or exposure—exerts a greater influence on seismic risk evolution? After a large event where vulnerability increases due to earthquake-induced damage, but the time-dependent hazard decreases, what are the resultant implications for seismic risk? (in the hypothetical example, the seismic risk increased after the Maule Earthquake, but this might not be the case in reality) How do cities grow, and how does this growth impact their seismic risk? The EASER project will result in an integrated methodology to systematically incorporate the time dependency of seismic hazard, vulnerability, and exposure, which will enhance our understanding of the dynamic nature of seismic risk.

The technological merit of the project resides in the EASER model, offering a comprehensive framework for assessing seismic risk while considering the temporal evolution of its elements. This model stands as a rational and transparent tool accessible to both public and private stakeholders, facilitating the evaluation of diverse seismic risk mitigation strategies. Notably, the hypothetical example in Figure 6 illustrates the model's utility, demonstrating that a stationary or a short-term assessment may favor a retrofit program, while a long-term evaluation reveals that updating seismic codes emerges as the more effective option. The EASER model thus empowers stakeholders to make informed decisions, contributing to the development of more resilient cities.

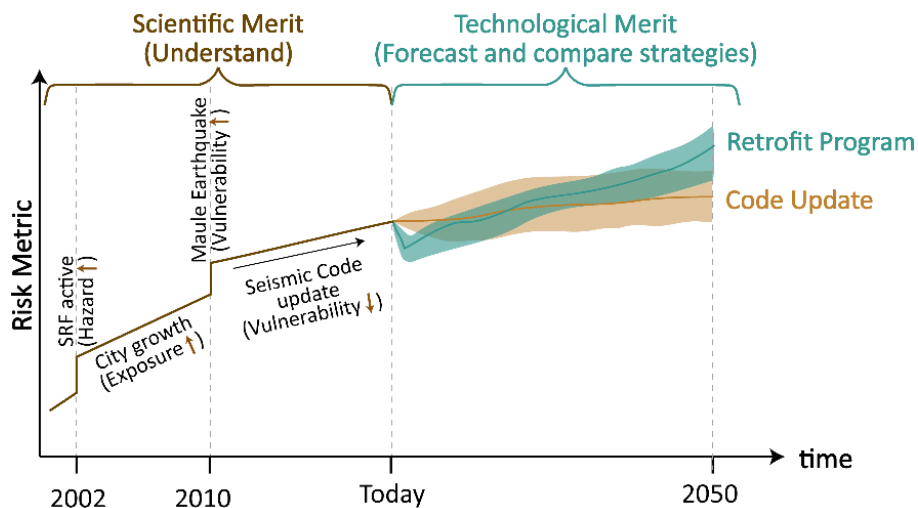


Figure 6. Hypothetical example of a seismic risk metric's evolution (e.g., expected number of displaced families) in Santiago, Chile. Shaded areas represent the aleatory uncertainty in the future risks. Both the past and future seismic risk will also have epistemic uncertainty, not illustrated herein.

b. Interested Institutions and Stakeholders

As part of the creation of SENAPRED (National Service for Disaster Prevention and Response), there are requirements related to the evaluation of natural risk with a primary focus on implementing measures to mitigate the associated losses. One of the main tasks corresponds to the comprehensive evaluation of seismic risk. For this purpose, they have developed a system to combine information from natural hazards and exposure, interactive and user-friendly, open to the public. This system is known as Visor Chile Preparado. This project aims to analyze the evolution of seismic risk in time, considering variations of hazard, exposition, and vulnerability. The scientific community has not investigated these concepts; hence, opening an opportunity for this research. As part of this collaboration, SENAPRED will provide relevant information from their database shown in the Visor Chile Preparado system, such as the distribution of relevant services (i.e., health, education, fire stations) and other critical infrastructures. On the other hand, the same system will help us in the dissemination of the seismic hazard estimations obtained in this project. Finally, the same system will enable the common user to examine the evolution of seismic risk estimations, improving the communications of these risks and helping to improve the preparedness.

The contribution of VMB, a private engineering consultancy, will be primarily focused on the components of exposure and seismic vulnerability. VMB has extensive experience in the structural analysis and design of buildings in Chile, including residential, office buildings, and industrial projects. VMB will contribute mainly in three aspects to the project: (i) characterization of the RC building typologies in terms of structural properties such as structural system, floor plan area, slenderness ratio, wall density ratio, among others; (ii) selection of a set of existing buildings with structural characteristics that are representative of the respective building typology, providing detailed information about such as drawings, calculation sheets and analytical models; and (iii) development of detailed FEM of each selected building and conducting NLTH analysis following Performance-Based Earthquake Engineering (PBEE) approach. We will work closely with VMB to identify the best repair and/or retrofit strategies to reduce the seismic risk of the buildings based on observed damage of critical structural elements and their extensive past experience in the field.

c. Technology Readiness Levels (TRL)

The research team for the project has made significant progress in domains such as multiscale geotechnical modeling for seismic safety and seismic engineering. They have developed non-ergodic seismic demand models and tools to assess liquefaction in subduction zones. In addition, researchers such as Gonzalo Montalva and Felipe Leyton have collaborated to obtain site parameters (shear wave velocities and site period) using surface wave techniques for seismic site classification. They have also conducted work on monitoring the structures of bridges. The contributions of Rosita Jünemann, Pablo Heresi, and Jorge Crempien in seismic risk analysis, the valuation of catastrophic risks, and the creation of technological platforms for visualizing risk results have complemented these efforts.

Previous research and projects funded by FONDEF have raised the existing technology to a Technology Readiness Level (TRL) 3, where an experimental proof-of-concept with basic computational models and software prototypes has been conducted. These methods have been validated in a laboratory/field-case setting to determine their viability in seismic risk analysis. Based on these findings, the project aims to reach Technology Readiness Level (TRL) 6 within three years. This will involve progressing from the experimental



proof of concept (TRL 3) to the demonstration of technology in a relevant environment (TRL 6). The advanced computational tools being developed will be implemented in real operational environments, albeit on a limited or controlled scale.

It is important to ensure that these tools meet all necessary operational and safety requirements for their application in urban planning, design, and retrofitting of infrastructure. This includes a comprehensive probabilistic analysis of seismic risk in the Andean subduction zone. By the end of the project, we expect the team members' applications to reflect a level of maturity that enables implementation in urban planning and infrastructure design projects under real conditions, albeit on a controlled scale. This will allow for the assessment of their effectiveness and efficiency before moving to more advanced stages of development and broader-scale deployment.

In summary, the current Technology Readiness Levels (TRLs) reflect the phase of experimental development and laboratory validation. The TRL expected at the end of the project indicates a phase of implementation and testing in environments that simulate real operational conditions. This sets the groundwork for future integrations and validations on a larger scale.

d. Research Lines

Research Line 1 – Time evolution of seismic hazard: Integrating statistical and physical methods to develop time dependent probabilistic models for seismic intensity provides valuable input for RL2 and RL4. Models will integrate products of the temporal dimension including data from paleoseismology (in collaboration with the group of Jasper Moernaut), spatial variations of the crust upper structure and segmentation (using part of the outcomes of previous research projects by the team), and shallow crustal seismogenic faults. This integration will allow the decision-making process to quantitatively account for the time-dependent probability that an earthquake of a given magnitude occurs in a particular region, with surface intensities that consider several aspects of the physical process that are currently neglected (i.e., considered uncertainties). This research line will be led by Felipe Leyton and Gonzalo Montalva. Additionally, it will include the participation of Jorge Crempien (Associate Researcher), one postdoctoral scholar (Postdoc1), and two MSc students (MSc1 during Years 1 and 2; MSc2 during Year 3). This research line will also be supported by international collaborations with Fabrice Cotton (GFZ Postdam), Graeme Weatherill (GFZ Postdam), Adrian Rodriguez-Marek (Virginia Polytechnic Institute and State University) and César Pastén (Universidad de Chile).

Research Line 2 – Time evolution of seismic vulnerability: This research line will investigate the temporal evolution of structural vulnerability and characterize it through analytical time-dependent fragility functions, particularly reinforced concrete and masonry Chilean building typologies. Key time-dependent factors within the Chilean context, such as cumulative earthquake damage and chloride- and carbonatation-induced corrosion will be evaluated. The research line involves selecting archetype buildings representative of prevalent building typologies, developing nonlinear analytical models with time-dependent effects, and validating them using experimental data collected from ambient vibration tests on existing structures. Analytical fragility functions, accounting for time-dependent effects, will be developed for selected buildings. The temporal evolution of fragility functions for each archetype will be analyzed, and correlations will be established between time-dependent and time-independent fragility functions across different building types. This research line will be led by Rosita Jünemann and Sebastián Calderón. It will include the participation of one PhD student (PhD1), two MSc students (MSc3 during Years 1 and 2; MSc4 during Year 3), and the support of Carlos Arteta (Universidad del Norte), Miguel Bravo-Haro (City, University of London), Cristian Cruz (Universidad Técnica Federico Santa María) and Gabriele Milani (Politecnico di Milano).

Research Line 3 – Time evolution of exposure: This research line will investigate the temporal evolution of urban exposure, focused on residential structures in the cities of Santiago, Viña del Mar, and Concepción. It will encompass aspects such as the number of residential structures, their locations, and various properties including age since construction, building height, etc. This research line will be split into four main tasks: (1) historical aerophotogrammetry and satellite image processing; (2) Gathering and analysis of publicly available building data; (3) development of enhanced time-dependent exposure model; and (4) validation experiment with historical data. This research line will be led by Pablo Heresi and Gonzalo Montalva. Additionally, it will include the participation of one postdoctoral scholar (Felipe Orellana, funding from ANID Postdoctoral Project #3230061) and the international collaboration with Vitor Silva (GEM Foundation).

Research Line 4 – Development of computational tools for evaluating time-dependent seismic risks: The final research line of EASER focuses on the technological aspect of the project and is divided into two primary tasks. The first task involves the development of a set of tools for the time-dependent evaluation of seismic risk, entitled EASER model. These tools will be curated and made accessible through public repositories. The second task entails using these tools to compare future seismic risk mitigation strategies in the cities of Santiago, Viña del Mar, and Concepción. These three cities are selected because of their significance for Chile in terms of both population and economy. Several risk mitigation strategies will be evaluated through the EASER models. This research line will be led by Rosita Jünemann and Pablo Heresi. It will include the participation of one PhD student (PhD2), and two MSc students (MSc5 during



Year 3; MSc6 during Year 3). This research line will also be supported by international collaborations with Eduardo Miranda (Stanford University), Graeme Weatherill (GFZ Postdam), and Vitor Silva (GEM Foundation).

e. Human capital training and formation

The EASER training and development plan is designed for the project to develop advanced computational tools for seismic risk assessment in the Andean subduction zone, along with training a new generation of earthquake scientists. The plan focuses on creating highly specialized human capital in earthquake related urban planning and infrastructure retrofitting, using a comprehensive analytical and probabilistic approach, with an unwavering commitment to gender equity in science and technology.

I. Training and Development Plan Objectives

- To train masters, doctoral, and postdoctoral researchers in advanced competencies in computational modeling and probabilistic seismic risk analysis.
- To ensure the practical application of research findings to enhance urban resilience in the Andean region.
- To balance gender representation within the project, actively promoting the participation of women at all levels of training.

II. Training and Development Strategies

- Participation of project researchers as advisors in graduate programs, providing specialized knowledge in seismic risk management.
- Development of tutorials directed at graduate students, oriented towards applying computational models in real scenarios.
- The design of one specialized course in seismic risk mitigation technologies for undergraduate and graduate programs, including practical workshops and simulations. This course will be available in the three participating universities (i.e., Main and Associate Institutions).
- The establishment of a mentoring program that connects students with industry and academia professionals, fostering knowledge and experience exchange.

III. Gender Perspective Action Plan

- The definition of clear policies for the inclusion of women in the research team, with specific participation goals.
- The creation of recruitment and promotion initiatives to attract women to study and work in fields related to seismic risk management.
- The implementation of flexible work policies that facilitate the reconciliation of work and personal life, especially for women in the team.
- The organization of motivational workshops for female students, especially during school stage, aiming to promote the discipline of earthquake engineering and the significant role women have in it.

IV. Results Transfer

- Integrating research outcomes into the training curriculum ensures that students work on immediately relevant applied problems.
- The promotion of collaboration with governmental (SENAPRED) and private entities to apply the knowledge and tools developed in seismic risk mitigation.
- The promotion of publication and dissemination of research results, demonstrating the effectiveness of the developed strategies and tools.

V. Our Commitment

To gather and train a group of researchers capable of significantly contributing to seismic risk management in the Andean subduction zone, with particular attention to gender equity and the transfer of technological knowledge to society, public entities responsible for territorial planning and management, and private stakeholders.

This training and development plan will be subject to periodic evaluations to ensure its alignment with the project objectives and effectiveness in achieving them. Continuous feedback from all participants will be sought to adapt and improve teaching and learning strategies and effective collaborations, thus ensuring academic excellence and the practical relevance of the training provided.

f. Collaboration

The participation of SENAPRED and VMB Structural Engineering in the "Evolution Assessment of Seismic Risk (EASER)" project is of high relevance due to its direct relation to the project's objectives and scope. Including these institutions not only reinforces the importance of the proposed technological development but also broadens the impact and applicability of the results.

SENAPRED's Involvement:

As a governmental entity focused on disaster management, SENAPRED is intrinsically connected to the central issue of the project: the evaluation and mitigation of seismic risk. They are mandated by law to provide hazard and risk maps to the country, products that are not readily available, and EASER will help develop them in close collaboration with SENAPRED staff.

Their experience in emergency planning and disaster response provides an essential perspective on the practical and operational needs in seismic risk management.

Contribution to Technological Development:

SENAPRED can provide historical and current data on earthquakes and their impacts, and government mitigation strategies, which are crucial for validating and calibrating seismic risk forecast tools. The agency can act as a key user of the developed tools, offering valuable feedback for their improvement and adaptation to real-world scenarios.

Insertion in Strategy and Activities:

SENAPRED is committed to participating in training and dissemination activities, aiding in transferring knowledge and the tools developed among communities and other relevant actors in disaster management. Their collaboration is critical in testing and validating the project products, ensuring that the technological developments are effectively applicable in emergency management.

VMB Structural Engineering's Involvement:

As a company specializing in structural engineering projects, VMB has a direct interest in accurately modeling seismic risk and in designing earthquake-resistant structures. Their structural analysis expertise complements the project's scientific and technological focus.

Contribution to Technological Development:

VMB can provide specialized technical knowledge in assessing structural vulnerability and in designing mitigation solutions, provide archetypes and help in the numerical modeling of real structural designs. The company can utilize the tools developed in the project to enhance its seismic engineering designs and consultancy services, a model which we expect other companies to follow.

Insertion in Strategy and Activities:

VMB can participate in the design and development phase of the computational tools, mainly in RL2 contributing their experience in the requirements and standards of the construction industry. The company can be a key partner in disseminating and applying the developed technologies in the private sector, especially in construction and civil engineering.

International and National Collaboration:

The largest centers in Europe dedicated to Earthquake Sciences, GEM Foundation and GFZ (German Research Center for Geosciences) are key partners with whom the research team has a long-standing relationship and are invested in EASER due to the scientific and practical relevance of the products sought. Also, some of the leading scientists in seismic risk (e.g., Prof. Miranda, who is one of the founders of the PEER PBEE framework for seismic risk assessment) and seismic hazard (i.e., Prof. Rodriguez-Marek) are also keen to collaborate with, and be part of, the EASER research team. They all have an interest in mentoring our students, publishing novel articles associated with the proposal, and contributing to the development of the EASER model.

The collaboration among the top three universities in Chile (in terms of students and faculty, see https://urapcenter.org/Rankings/2023-2024/World_Ranking_2023-2024) along with the engagement of national collaborators and international partners, will foster a dynamic and enriching academic environment. It is important to highlight that the research team assembled, complemented by our international collaborators, stands as the leading group in Chile uniquely prepared to conduct a research project of this magnitude concerning seismic risk assessment. This collaborative framework will significantly enhance academic productivity and play a pivotal role in supporting the development and training of young scientists. The diverse perspectives and expertise brought together through this collaborative endeavor are anticipated to contribute not only to the project's success but also to the broader academic advancement in the field of earthquake engineering.

Finally, collaborating with SENAPRED and VMB Structural Engineering is strategically pertinent and beneficial. SENAPRED, focusing on disaster prevention and response, and VMB, specializing in structural engineering, provide invaluable knowledge and capabilities to the success of this endeavor. This collaboration brings together both public and private sectors, enriching the development and application of the project. Both institutions play critical roles in the validation, implementation, and dissemination of the developed technological solutions, ensuring that the project is not only scientifically robust but also practically applicable and aligned with the needs of the public and private sectors.

g. Extension and Dissemination of Results

The topics related to seismic risk should be comprehensively addressed, reaching both professionals and the non-specialized public through specific strategies targeted at academic and non-academic audiences alike. The knowledge transfer and dissemination activities will be divided into two large groups, **(1) Towards a non-specialized and/or scientific public** and **(2) Towards a specialized**

and/or scientific public. Additionally, there will be a project journalist to ensure the correct execution of the dissemination activities and the results will be communicated to the government entities responsible for seismic hazard in Chile (SENAPRED, SHOA, SERNAGEOMIN, CSN), starting by our partner SENAPRED with whom we have the agreement to work closely.

TASK ED1. Non-specialized or scientific public

A gender focus will be considered, seeking to reach, above all, women in secondary and university education to reduce the gender gap in STEM careers, for which we will collaborate with non-profit organizations such as "Niñas Pro" through online talks and face-to-face workshops. This initiative will be carried out throughout the three years of the project as the participating students change from year to year. On the other hand, the project will collaborate with speakers on podcast platforms such as "La Ciencia Pop" to inform about the discipline, curiosities, and updates in the area in a more friendly way. In addition, didactic material will be developed to help in the teaching of seismic risk, mainly for high school and university students. Finally, the project will be completed with a face-to-face activity at a university. This will be a science fair where seismic risk will be taught through playful activities, short talks, and workshops, aimed at a non-specialized public.

TASK ED2. Specialized and/or scientific public

The results obtained during project implementation will be published. In addition, the results will be presented at national and international conferences and seminars such as GIASIS, SSA and EERI annual meeting, all activities that are carried out every year. The results obtained will be incorporated into SENAPRED's "Chile Prepared" portal; and, once the project software is completed, training workshops will be organized for decision making institutions such as SENAPRED, SERNAGEOMIN, SHOA and CSN, as well as private and nonprofit organizations interested in the generated products such as AICE (Structural engineers association), REDMIC (women civil engineers network), SOCHIGE (Geotechnical National Society), or ACHISINA (Earthquake engineer association).

h. Application and/or Knowledge Transfer and Impact in other (Non Academic) Sectors

The Research and Development Team's Experience in Knowledge and Technology Transfer

The research and development team has an outstanding track record in the transfer of knowledge and technology to both public and private productive sectors. Team members, including Gonzalo Montalva, Sebastián Calderón, Felipe Leyton, Pablo Heresi, and Rosita Jünemann, have participated or led more than 45 research projects ranging from seismic hazard assessments tools to the development to earthquake risk implementations. In addition, the team has led the way in ground motion prediction for site-specific earthquake design. As a result, the team has participated, and/or their tools been used, in large mining projects, energy developments, and seismic safety evaluations of major hospitals and transportation infrastructure, to name the most relevant sectors. Their work has transcended academia, directly impacting the industry through specialized models, consultancies and advisories, and contributing to the creation of policies and regulations that improve seismic resilience.

Transfer Strategy for EASER

The transfer strategy for this project is structured around the following key activities:

- **Collaboration with Government Entities and Private Sector Organizations:** We will establish formal partnerships with SENAPRED and VMB Structural Engineering to integrate the developed models and tools into existing systems and practices. Collaborations will enable the validation of technologies in real environments and their adaptation to the specific needs of end-users. The team will use its current involvement with national societies such as the National Geotechnical Soc. (SOCHIGE), Earthquake Assoc. (ACHISINA), or Structural Eng. Assoc. (AICE) to promote the project's results as well as the formal EASER partners.
- **Participation in Conferences and Sectorial Events:** Researchers will actively participate in national and international conferences to present the progress of the project and establish contacts with potential collaborators and users of the developed technologies. The team will host at least two special sessions at such events.
- **Training and Awareness:** Workshops and talks will be organized at universities, research centers, and communities, focusing on the importance of seismic engineering and the role of women in science and technology. These activities will include the creation of educational and informative materials, as well as participation in scientific fairs and events.
- **Continuous Feedback and Technology Improvement:** Through exchanges and internships with our international partners and the realization of annual workshops with stakeholders, feedback will be collected to continuously improve the developed tools.

This strategy highlights the importance of the practical application and transfer of developed technologies beyond academic confines, aiming to contribute to Chile's economic development and social welfare. The collaborative and participative approach ensures that the developed solutions are relevant, applicable, and sustainable, positively impacting the country's seismic risk management and structural safety. Potential impacts beyond Chile are also expected and will be sought.

- Bibliographic References:** This section is filled in the file BIBLIOGRAPHIC-REFERENCES-2024.docx, which is uploaded in the online application system.

j. Indicators

In this section you will find a list with mandatory indicators and boxes in blank for specific metrics to show results of this project. The specific indicators will be used to measure the quality and potential impact of the project and must be defined precisely and consistent with the commitments established in the proposal. In addition, the methodology that will be used to measure each indicator is required to be clearly described, specifying how data will be collected, how evaluations will be carried out, and how results will be quantified.

SCIENTIFIC EXCELLENCE	Baseline*	Year 1	Year 2	Year 3
Number of WoS publications ¹	14.7	15	20	25
Number of non WoS publications ²	2.3	3	5	8
Percentage of publications Co-authored with researchers of the Project ³	24.7%	25	30	50
Number of national presentations/conferences/ industry relations instances ⁴	6.3	6	12	18
Number of international presentations/conferences ⁵	9	10	10	10
Others (Describe and add calculation method)				

Notes:

* Average of last 3 years, considering productivity of Principal and Associate Researchers.

¹ Only considers indexed publications related to the topics or lines of research of the project.

² Only considers non indexed publications related to the topics or lines of research of the project.

³ Only considers publications related to topics or research lines of the project, whose authors are Principal and/or Associate Researchers from this project.

^{4,5} Considers instances of participation and/or organization.

TECHNOLOGICAL TRANSFER AND LINKAGE WITH OTHER SECTORS OF CHILEAN ECONOMY AND SOCIETY	Baseline*	Year 1	Year 2	Year 3
Number of patent applications ¹	0	0	0	0
Number of patent application's granted ²	0	0	0	0
Number of licenses and/or Technology Transfer Agreements applied and/or granted	1	0	0	1
Number of spin-offs companies rising from the project ³	0	0	0	0
N° of associative projects with enterprises	2.7	0	1	2
N° of agreements with the industrial sector / enterprises ⁴	0.3	2	1	3
Others (please describe and add calculation method)				

Notes:

* Average of last 3 years, considering productivity of Principal and Associate Researchers.

¹ Considers all applications, approved, rejected and in process.

² Only considers approved patent.

³ Considers formalized spinoffs.

⁴ Refers to contracts or agreements for the execution of one or more projects or other types of collaboration formalized through agreements.



TRAINING OF HUMAN RESOURCES	Baseline*	Year 1	Year 2	Year 3
Number of Master students ¹	9	9	12	15
Number of completed master theses ²	4.3	4	5	6
Number of Ph.D. students ³	5.3	5	6	6
Number of completed Ph.D. theses ⁴	0.3	2	4	6
Number of posdocs working in the project	0.3	1	1	1
Number of undergraduate students ⁵	8	8	10	15
Others (please describe and add calculation method)				

Notes:

* Average of last 3 year for the project team.

^{1,3,5} Refers to students trained under the project.

^{2,4} Only considers theses on topics related to the project's research lines.

INTERNATIONAL NETWORKS	Baseline*	Year 1	Year 2	Year 3
Number of conjoint projects developed ¹	1	1	2	2
Percentage of ISI publications co-authored with international researchers (from other institutions) ²	50	50	50	50
Number of stays/visits from students or researchers from other Centers or projects (national and international)	1.7	2	3	4
Number of stays/visits to other institutions by students or researchers of the project	1	2	3	4
Others (please describe and add calculation method)				

Notes:

* Average of last 3 years, considering productivity of Principal and Associate Researchers.

¹ Considers all R&D projects in which the team members participate with other research groups, companies or international centers.

² Only considers publications related to the topics or lines of research of the project.

EXTENSION AND DISSEMINATION OF THE RESULTS	Baseline*	Year 1	Year 2	Year 3
Number of times the project appears in mass media	4	4	6	6
Number of documents, reports, proceedings resulting from outreach/extramural events or activities	1	2	2	3
Number of outreach and/or extension activities ¹	7	7	10	12
Total number of people attending to outreach and/or extension activities ²	70	70	200	300
Number of conjoint projects developed for the purpose of executing outreach activities	0	1	1	1
Others (please describe and add calculation method)				

Notes:

* Average of last 3 year for the project team.

¹ Considers activities or instances organized by the project.

² Considers attendees on site and virtual activities.



GENDER	Baseline*	Year 1	Year 2	Year 3
Percentage of female researchers in the project ¹	15	15	20	25
Percentage of women in the "principal researcher" category in the project ²	20	20	20	20
Number of publications with women among the authors ³	7	8	10	15
Number of graduate seminars for female students ⁴	1	2	3	4
Number of trainings on gender-related topics ⁵	0	0	0	0
Number of female postdoctoral fellows	0	0	1	1
Others (please describe and add calculation method)				

Notes:

* Average of last 3 year for the project team.

^{1,2} Of the total number of researchers in the project

³ Considers women researchers from the project and women researchers from other research groups.

⁴ Only considers seminars organized by the Research Team

⁵ Only considers the training activities organized by the Research Team.

V. WORKING PLAN

Objectives, Tasks, and Activities		Year 1 (Months)											
		1	2	3	4	5	6	7	8	9	10	11	12
OE1 - Develop time-dependent models of seismic hazard within the Andean subduction context, considering the time between earthquakes, the temporal evolution of locking degrees at the source, of path characteristics, and of site response effects													
TASK 1.1. Evolution of Non-Poissonian and Poissonian processes	Activity 1.1.1. Seismic hazard analysis with classical Poissonian approach												
	Activity 1.1.2. Exhaustive survey of historical and paleoseismological data related to earthquakes in the Andean subduction zone, both for seismicity associated with the subduction and the cortical regime												
	Activity 1.1.3. Study the frequency, magnitude and location of past earthquakes												
	Activity 1.1.4. Identification of aftershock and foreshock sequences, as well as other seismic crisis, and analysis of its effect in bvalue estimations												
	Activity 1.1.5. Conduct a time-dependent analysis using different statistical distributions (e.g., BPT, Gamma, Weibul, Lognormal)												
	Activity 1.1.6. Develop a non-time-dependent analysis of seismic hazard												
	Activity 1.1.7. Develop and use computational models to simulate earthquakes and their effects on different soil types to predict the temporal evolution of the hazard												
TASK 1.2. Evolution of site characteristics	Activity 1.2.1. Simultaneous modelling of source, attenuation, and site effects with non-parametric approach												
	Activity 1.2.2. Evaluation of time changes of source, attenuation, and site effects considering variable time windows												
	Activity 1.2.3. Repeat characterization of previously visited site to evaluate changes with time												
	Activity 1.2.4. Analysis of time changes in site characteristics of target cities												
TASK 1.3. Evolution of knowledge	Activity 1.3.1. Study and account the uncertainty on each seismic hazard analysis component												
	Activity 1.3.2. Consider and propagate appropriately the uncertainties on probabilistic seismic hazard analysis												
	Activity 1.3.3. Estimate the spatial and temporal evolution of the variation on strong ground motion intensities on Chilean subduction zone to conduct a partially- or non-ergodic analyses												
	Activity 1.3.4. Use case histories (i.e., Maule 2010, Iquique 2014, and Illapel 2015 megathrust) to evaluate the evolution of epistemic uncertainty and their influence on seismic hazard estimates												
	Activity 1.3.5. Generate a model to estimate a new epistemic uncertainty (space and time-dependent) and their impact to updating the seismic hazard on Concepción, Viña del Mar, and Santiago for connecting with risk and liquefaction assessment												
	Activity 1.3.6. Compute state parameter from CPT at the 7 locations in Concepción and evaluate the spatial dependence												
TASK 1.4. Earthquake-induced liquefaction	Activity 1.4.1. Compute shear-wave velocity 3D field at 4 of these locations and evaluate time evolution of these by using ORSCA (the site response observatory at the Concepción basin) and comparing with published data by the												
	Activity 1.4.2. Generate a model, and a computational tool, to estimate the magnitude and the uncertainty of expected liquefaction induced settlement given different kinds of data and considering time variation throughout												
	Activity 1.4.3. Compute state parameter from CPT at the 7 locations in Concepción and evaluate the spatial dependence												
	Activity 1.4.4. Compute shear-wave velocity 3D field at 4 of these locations and evaluate time evolution of these by using ORSCA (the site response observatory at the Concepción basin) and comparing with published data by the												
OE2 - Evaluate the evolution of structural vulnerability over time due to degradation of materials and occurrence of earthquake sequences, both from analytical and experimental approaches													
TASK 2.1. Archetype building characterization	Activity 2.1.1. Characterization of Masonry building typologies												
	Activity 2.1.2. Characterization of RC building typologies												
	Activity 2.1.3. Selection of Masonry archetypes												
	Activity 2.1.7. Selection of RC archetypes												
TASK 2.2. Development of detailed analytical models including time-dependent effects	Activity 2.2.1. FEM modelling of Masonry archetypes												
	Activity 2.2.2. FEM modelling of RC archetypes												
	Activity 2.2.3. Definition of sequential records												
	Activity 2.2.4. Assessment of degradation of mechanical properties due to previous earthquake damage												
	Activity 2.2.5. Assessment of degradation of mechanical properties of masonry due to corrosion												
	Activity 2.2.6. Assessment of degradation of mechanical properties of RC due to corrosion												
	Activity 2.2.7. Push-over and NLTH analyses of FEM models of masonry archetypes												
	Activity 2.2.8. Push-over and NLTH analyses of FEM models of RC archetypes												
	Activity 2.2.9. Identification of suitable retrofit options for masonry archetypes												
	Activity 2.2.10. Identification of suitable retrofit options for RC archetypes												
TASK 2.3. Experimental characterization of time-dependent degradation	Activity 2.3.1. Execution of Free vibration tests of Masonry archetypes												
	Activity 2.3.2. Execution of Free vibration tests of RC study archetypes												
	Activity 2.3.4. Analysis of Free vibration test results of Masonry archetypes												
	Activity 2.3.4. Analysis of Free vibration test results of RC archetypes												
	Activity 2.3.5. Analysis of projected and measured dynamic properties on archetypes												
	Activity 2.3.6. Analysis of time-dependent effects on archetypes from experimental data												
TASK 2.4. Development of time-dependent fragility functions	Activity 2.4.1. Simplified models of masonry archetypes												
	Activity 2.4.2. Simplified models of RC archetypes												
	Activity 2.4.3. Seismic record selection for fragility analysis												
	Activity 2.4.4. Development of NLTHA for building archetypes												
	Activity 2.4.5. Development of fragility functions												
	Activity 2.4.6. Analysis of temporal evolution of fragility functions												
OE3 - Assess the temporal evolution of exposure through analysis of publicly available databases and image recognition techniques applied to multispectral satellite imagery													
TASK 3.1. Historical aerophotogrammetry and satellite image processing	Activity 3.1.1. Data collect (Satellite imagery, ortophotos, Lidar and Building height)												
	Activity 3.1.2. Satellite image processing												
	Activity 3.1.3. Map generation over time exposure from multispectral and SAR data												
TASK 3.2. Gathering and analysis of publicly available building data	Activity 3.2.1. Data collect (cadastres SII)												
	Activity 3.2.2. Processing database and geolocate construction units												
	Activity 3.2.3. Estimate number of stories of each building												
TASK 3.3. Development of enhanced time-dependent	Activity 3.3.1. Combine models of Tasks 3.1 and 3.2												
TASK 3.4. Validation experiment with historical data	Activity 3.3.2. Develop time-dependent model to forecast future exposure												
	Activity 3.4.1. Identification of required information to forecast building exposure from 1985 to target date												
	Activity 3.4.2. Forecast the current exposure from 1985 to target date												
	Activity 3.4.3. Comparison of real and forecasted building exposure												
OE4 - Develop a set of computational tools to evaluate the effect of time-dependent seismic hazard, structural vulnerability and exposure on seismic risk, considering and propagating their uncertainties. We will use three cities within Chile as case studies for our tool set: Santiago, Viña del Mar, and Concepción													
TASK 4.1. Development of computational tools for	Activity 4.1.1. Curate and upload into repositories the computational tools developed before												
	Activity 4.1.2. Extend static formulation of seismic risk assessment to EASER model												
	Activity 4.2.1. Static seismic risk assessment of Santiago, Viña del Mar, and Concepción												
TASK 4.2. Test developed tools in case study cities	Activity 4.2.2. Time-dependent seismic risk assessment of Santiago, Viña del Mar, and Concepción												
	Activity 4.2.3. Evaluate seismic risk mitigation strategies in Santiago, Viña del Mar, and Concepción												
Extension and Dissemination of Results													
TASK ED1. Non-specialized or scientific public	Activity ED1.1. Talks for the organization Niñas pro, and one presentational workshop.												
	Activity ED1.2. Science podcast chapters in collaboration with Speakers.												
	Activity ED1.3. Production of content for social networks (infographics and reels).												
	Activity ED1.4. Web page creation and updates.												
	Activity ED1.5. Preparation of didactic material for teaching seismic risk.												
	Activity ED1.6. Scientific outreach fair seismic risk.												
TASK ED2. Specialized and/or scientific public	Activity ED2.1. Attendance at seminars and congresses.												



Objectives, Tasks, and Activities		Year 2 (Months)											
		1	2	3	4	5	6	7	8	9	10	11	12
OE1 - Develop time-dependent models of seismic hazard within the Andean subduction context, considering the time between earthquakes, the temporal evolution of locking degrees at the source, of path characteristics, and of site response effects													
TASK 1.1. Evaluation of Non-Poissonian and Poissonian processes	Activity 1.1.1. Seismic hazard analysis with classical Poissonian approach												
	Activity 1.1.2. Exhaustive survey of historical and paleoseismological data related to earthquakes in the Andean subduction zone, both for seismicity associated with the subduction and the cortical regime												
	Activity 1.1.3. Study the frequency, magnitude and location of past earthquakes												
	Activity 1.1.4. Identification of aftershock and foreshock sequences, as well as other seismic crisis, and analysis of its effect in bvalue estimations												
	Activity 1.1.5. Conduct a time-dependent analysis using different statistical distributions (e.g., BPT, Gamma, Weibul, Lognormal)												
	Activity 1.1.6. Develop a non-time-dependent analysis of seismic hazard												
TASK 1.2. Evolution of site characteristics	Activity 1.1.7. Develop and use computational models to simulate earthquakes and their effects on different soil types to predict the temporal evolution of the hazard												
	Activity 1.2.1. Simultaneous modelling of source, attenuation, and site effects with non-parametric approach												
	Activity 1.2.2. Evaluation of time changes of source, attenuation, and site effects considering variable time windows												
TASK 1.2. Evolution of site characteristics	Activity 1.2.3. Repeat characterization of previously visited site to evaluate changes with time												
	Activity 1.2.4. Analysis of time changes in site characteristics of target cities												
	Activity 1.3.1. Study and account the uncertainty on each seismic hazard analysis component												
	Activity 1.3.2. Consider and propagate appropriately the uncertainties on probabilistic seismic hazard ana analysis												
	Activity 1.3.3. Estimate the spatial and temporal evolution of the variation on strong ground motion intensities on Chilean subduction zone to conduct a partially- or non-ergodic analyses												
	Activity 1.3.4. Use case histories (i.e., Maule 2010, Iquique 2014, and Illapel 2015 megathrust) to evaluate the evolution of epistemic uncertainty and their influence on seismic hazard estimates												
TASK 1.2. Evolution of site characteristics	Activity 1.3.5. Generate a model to estimate a new epistemic uncertainty (space and time-dependent) and their impact to updating the seismic hazard on Concepción, Viña del Mar, and Santiago for connecting with risk and liquefaction assessment												
	Activity 1.4.1. Compute state parameter from CPT at the 7 locations in Concepción and evaluate the spatial dependence												
	Activity 1.4.2. Compute shear-wave velocity 3D field at 4 of these locations an evaluate time evolution of these by using ORSCA (the site response observatory at the Concepción basin) and comparing with published data by the												
OE2 - Evaluate the evolution of structural vulnerability over time due to degradation of materials and occurrence of earthquake sequences, both from analytical and experimental approaches													
TASK 2.1. Archetype building characterization	Activity 2.1.1. Characterization of Masonry building typologies												
	Activity 2.1.2. Characterization of RC building typologies												
	Activity 2.1.3. Selection of Masonry archetypes												
	Activity 2.1.7. Selection of RC archetypes												
TASK 2.2. Development of detailed analytical models including time-dependent effects	Activity 2.2.1. FEM modelling of Masonry archetypes												
	Activity 2.2.2. FEM modelling of RC archetypes												
	Activity 2.2.3. Definition of sequential records												
	Activity 2.2.4. Assessment of degradation of mechanical properties due to previous earthquake damage												
	Activity 2.2.5. Assessment of degradation of mechanical properties of masonry due to corrosion												
	Activity 2.2.6. Assessment of degradation of mechanical properties of RC due to corrosion												
TASK 2.3. Experimental characterization of time-dependent degradation	Activity 2.2.7. Push-over and NLTH analyses of FEM models of masonry archetypes												
	Activity 2.2.8. Push-over and NLTH analyses of FEM models of RC archetypes												
	Activity 2.2.9. Identification of suitable retrofit options for masonry archetypes												
	Activity 2.2.10. Identification of suitable retrofit options for RC archetypes												
	Activity 2.3.1. Execution of Free vibration tests of Masonry archetypes												
	Activity 2.3.2. Execution of Free vibration tests of RC study archetypes												
TASK 2.4. Development of time-dependent fragility functions	Activity 2.3.4. Analysis of Free vibration test results of Masonry archetypes												
	Activity 2.3.4. Analysis of Free vibration test results of RC archetypes												
	Activity 2.3.5. Analysis of projected and measured dynamic properties on archetypes												
	Activity 2.3.6. Analysis of time-dependent effects on archetypes from experimental data												
OE3 - Assess the temporal evolution of exposure through analysis of publicly available databases and image recognition techniques applied to multispectral satellite imagery													
TASK 3.1. Historical aerophotogrammetry and satellite image processing	Activity 3.1.1. Data collect (Satellite imagery, ortophotos, Lidar and Building height)												
	Activity 3.1.2. Satellite image processing												
	Activity 3.1.3. Map generation over time exposure from multispectral and SAR data												
TASK 3.2. Gathering and analysis of publicly available building data	Activity 3.2.1. Data collect (cadastres SII)												
	Activity 3.2.2. Processing database and geolocate construction units												
	Activity 3.2.3. Estimate number of stories of each building												
TASK 3.3. Development of enhanced time-dependent	Activity 3.3.1. Combine models of Tasks 3.1 and 3.2												
	Activity 3.3.2. Develop time-dependent model to forecast future exposure												
TASK 3.4. Validation experiment with historical data	Activity 3.4.1. Identification of required information to forecast building exposure from 1985 to target date												
	Activity 3.4.2. Forecast the current exposure from 1985 to targed date												
	Activity 3.4.3. Comparison of real and forecasted building exposure												
OE4 - Develop a set of computational tools to evaluate the effect of time-dependent seismic hazard, structural vulnerability and exposure on seismic risk, considering and propagating their uncertainties. We will use three cities within Chile as case studies for our tool set: Santiago, Viña del Mar, and Concepción													
TASK 4.1. Development of computational tools for	Activity 4.1.1. Curate and upload into repositories the computational tools developed before												
	Activity 4.1.2. Extend static formulation of seismic risk assessment to EASER model												
TASK 4.2. Test developed tools in case study cities	Activity 4.2.1. Static seismic risk assessment of Santiago, Viña del Mar, and Concepción												
	Activity 4.2.2. Time-dependent seismic risk assessment of Santiago, Viña del Mar, and Concepción												
Extension and Dissemination of Results													
TASK ED1. Non-specialized or scientific public	Activity ED1.1. Talks for the organization Niñas pro, and one presential workshop.												
	Activity ED1.2. Science podcast chapters in collaboration with Speakers.												
	Activity ED1.3. Production of content for social networks (infographics and reels).												
	Activity ED1.4. Web page creation and updates.												
	Activity ED1.5. Preparation of didactic material for teaching seismic risk.												
	Activity ED1.6. Scientific outreach fair seismic risk.												
TASK ED2. Specialized and/or scientific public	Activity ED2.1. Attendance at seminars and congresses.												

Objectives, Tasks, and Activities		Year 3 (Months)											
		1	2	3	4	5	6	7	8	9	10	11	12
OE1 - Develop time-dependent models of seismic hazard within the Andean subduction context, considering the time between earthquakes, the temporal evolution of locking degrees at the source, of path characteristics, and of site response effects													
TASK 1.1. Evaluation of Non-Poissonian and Poissonian processes	Activity 1.1.1. Seismic hazard analysis with classical Poissonian approach												
	Activity 1.1.2. Exhaustive survey of historical and paleoseismological data related to earthquakes in the Andean subduction zone, both for seismicity associated with the subduction and the cortical regime												
	Activity 1.1.3. Study the frequency, magnitude and location of past earthquakes												
	Activity 1.1.4. Identification of aftershock and foreshock sequences, as well as other seismic crisis, and analysis of its effect in bvalue estimations												
	Activity 1.1.5. Conduct a time-dependent analysis using different statistical distributions (e.g., BPT, Gamma, Weibul, Lognormal)												
	Activity 1.1.6. Develop a non-time-dependent analysis of seismic hazard												
	Activity 1.1.7. Develop and use computational models to simulate earthquakes and their effects on different soil types to predict the temporal evolution of the hazard												
TASK 1.2. Evolution of site characteristics	Activity 1.2.1. Simultaneous modelling of source, attenuation, and site effects with non-parametric approach												
	Activity 1.2.2. Evaluation of time changes of source, attenuation, and site effects considering variable time windows												
	Activity 1.2.3. Repeat characterization of previously visited site to evaluate changes with time												
TASK 1.2. Evolution of site characteristics	Activity 1.2.4. Analysis of time changes in site characteristics of target cities												
	Activity 1.3.1. Study and account the uncertainty on each seismic hazard analysis component												
	Activity 1.3.2. Consider and propagate appropriately the uncertainties on probabilistic seismic hazard ana analysis												
	Activity 1.3.3. Estimate the spatial and temporal evolution of the variation on strong ground motion intensities on Chilean subduction zone to conduct a partially- or non-ergodic analyses												
	Activity 1.3.4. Use case histories (i.e., Maule 2010, Iquique 2014, and Illapel 2015 megathrust) to evaluate the evolution of epistemic uncertainty and their influence on seismic hazard estimates												
TASK 1.2. Evolution of site characteristics	Activity 1.3.5. Generate a model to estimate a new epistemic uncertainty (space and time-dependent) and their impact to updating the seismic hazard on Concepción, Viña del Mar, and Santiago for connecting with risk and liquefaction assessment												
	Activity 1.4.1. Compute state parameter from CPT at the 7 locations in Concepción and evaluate the spatial dependence												
	Activity 1.4.2. Compute shear-wave velocity 3D field at 4 of these locations an evaluate time evolution of these by using ORSCA (the site response observatory at the Concepción basin) and comparing with published data by the												
	Activity 1.4.3. Generate a model, and a computational tool, to estimate the magnitude and the uncertainty of expected liquefaction induced settlement given different kinds of data and considering time variation throughout												
OE2 - Evaluate the evolution of structural vulnerability over time due to degradation of materials and occurrence of earthquake sequences, both from analytical and experimental approaches													
TASK 2.1. Archetype building characterization	Activity 2.1.1. Characterization of Masonry building typologies												
	Activity 2.1.2. Characterization of RC building typologies												
	Activity 2.1.3. Selection of Masonry archetypes												
	Activity 2.1.7. Selection of RC archetypes												
TASK 2.2. Development of detailed analytical models including time-dependent effects	Activity 2.2.1. FEM modelling of Masonry archetypes												
	Activity 2.2.2. FEM modelling of RC archetypes												
	Activity 2.2.3. Definition of sequential records												
	Activity 2.2.4. Assessment of degradation of mechanical properties due to previous earthquake damage												
	Activity 2.2.5. Assessment of degradation of mechanical properties of masonry due to corrosion												
	Activity 2.2.6. Assessment of degradation of mechanical properties of RC due to corrosion												
	Activity 2.2.7. Push-over and NLTH analyses of FEM models of masonry archetypes												
TASK 2.3. Experimental characterization of time-dependent degradation	Activity 2.2.8. Push-over and NLTH analyses of FEM models of RC archetypes												
	Activity 2.2.9. Identification of suitable retrofit options for masonry archetypes												
	Activity 2.2.10. Identification of suitable retrofit options for RC archetypes												
	Activity 2.3.1. Execution of Free vibration tests of Masonry archetypes												
	Activity 2.3.2. Execution of Free vibration tests of RC study archetypes												
	Activity 2.3.4. Analysis of Free vibration test results of Masonry archetypes												
	Activity 2.3.4. Analysis of Free vibration test results of RC archetypes												
TASK 2.4. Development of time-dependent fragility functions	Activity 2.3.5. Analysis of projected and measured dynamic properties on archetypes												
	Activity 2.3.6. Analysis of time-dependent effects on archetypes from experimental data												
	Activity 2.4.1. Simplified models of masonry archetypes												
	Activity 2.4.2. Simplified models of RC archetypes												
	Activity 2.4.3. Seismic record selection for fragility analysis												
	Activity 2.4.4. Development of NLTHA for building archetypes												
	Activity 2.4.5. Development of fragility functions												
OE3 - Assess the temporal evolution of exposure through analysis of publicly available databases and image recognition techniques applied to multispectral satellite imagery	Activity 2.4.6. Analysis of temporal evolution of fragility functions												
	Activity 3.1.1. Data collect (Satellite imagery, ortophotos, Lidar and Building height)												
	Activity 3.1.2. Satellite image processing												
	Activity 3.1.3. Map generation over time exposure from multispectral and SAR data												
	Activity 3.2.1. Data collect (cadastres SII)												
	Activity 3.2.2. Processing database and geolocate construction units												
	Activity 3.2.3. Estimate number of stories of each building												
TASK 3.3. Development of enhanced time-dependent	Activity 3.3.1. Combine models of Tasks 3.1 and 3.2												
	Activity 3.3.2. Develop time-dependent model to forecast future exposure												
	Activity 3.4.1. Identification of required information to forecast building exposure from 1985 to target date												
	Activity 3.4.2. Forecast the current exposure from 1985 to targed date												
TASK 3.4. Validation experiment with historical data	Activity 3.4.3. Comparison of real and forecasted building exposure												
OE4 - Develop a set of computational tools to evaluate the effect of time-dependent seismic hazard, structural vulnerability and exposure on seismic risk, considering and propagating their uncertainties. We will use three cities within Chile as case studies for our tool set: Santiago, Viña del Mar, and Concepción													
TASK 4.1. Development of computational tools for	Activity 4.1.1. Curate and upload into repositories the computational tools developed before												
	Activity 4.1.2. Extend static formulation of seismic risk assessment to EASER model												
TASK 4.2. Test developed tools in case study cities	Activity 4.2.1. Static seismic risk assessment of Santiago, Viña del Mar, and Concepción												
	Activity 4.2.2. Time-dependent seismic risk assessment of Santiago, Viña del Mar, and Concepción												
	Activity 4.2.3. Evaluate seismic risk mitigation strategies in Santiago, Viña del Mar, and Concepción												
Extension and Dissemination of Results													
TASK ED1. Non-specialized or scientific public	Activity ED1.1. Talks for the organization Niñas pro, and one presential workshop.												
	Activity ED1.2. Science podcast chapters in collaboration with Speakers.												
	Activity ED1.3. Production of content for social networks (infographics and reels).												
	Activity ED1.4. Web page creation and updates.												
	Activity ED1.5. Preparation of didactic material for teaching seismic risk.												
	Activity ED1.6. Scientific outreach fair seismic risk.												
TASK ED2. Specialized and/or scientific public	Activity ED2.1. Attendance at seminars and congresses.												

Possible barriers or technical challenges to achieving the project outcomes include a lack of interested students; not enough dedicated personnel by the interested institutions; or lack of optimal satellite imagery. For the lack of students, although we expect the opposite, we have money allocated in the budget to incentivise their involvement, and we have already enrolled two PhD students (Catalina Cabello and Nicolás Bastías) plus a postdoctoral scholar who is already funded (Dr. Felipe Orellana), one of the PhDs is also already funded by ANID, together with a strong research team. For the second barrier, the research team could adapt to working with other public institutions such as SERNAGEOMIN (Mining and Geology Service) or Ministry of Public work, both entities with whom we have already worked in the past; SENAPRED is the ideal entity to work with in this topic and we do not foresee a lack of commitment on their part, the opposite in fact. Finally, for the unlikely case of lack of optimal satellite imagery, we could adapt to the use of mobile-phone data more intensively and freely available imagery from Sentinel 2.

VI. RESEARCH TEAM ORGANIZATION

Figure 7 presents the organization chart of the project, which shows the relationship between all the participants. As it can be seen, there are two interested institutions: SENAPRED and VMB Ingeniería Estructural S.A. Three Chilean universities participate in the project. The main institution will be the Universidad de Concepción, while Universidad de Chile and Pontificia Universidad Católica de Chile will be the associate institutions. From the main institution, two researchers will be involved: Gonzalo Montalva and Sebastián Calderón; who will be the director and deputy director of the project, respectively. Two Pontificia Universidad Católica de Chile academics will participate, Rosita Jünemann as PI and Jorge Crempien as associate researcher. Two researchers from the Universidad de Chile, Felipe Leyton and Pablo Heresi, will contribute as PI's. The six academics will supervise postdoctorates (2), Ph.D. (2), and M.Sc. (6) students, whose expected advisors are shown in the Figure 7, and the rest of the team will serve in their committees. The project also considers the contribution of prestigious collaborators from abroad (U.S., Germany, Italy, England, Austria, and Colombia) and Chile (Universidad de Chile and Universidad Técnica Federico Santa María), who are listed in the Figure 7. The project also considers personnel for administrative and, linkage and communication activities. Furthermore, the collaboration of more students and academics is expected, although they are not individualized in the chart since they will be contacted and involved in the project for as long as required.

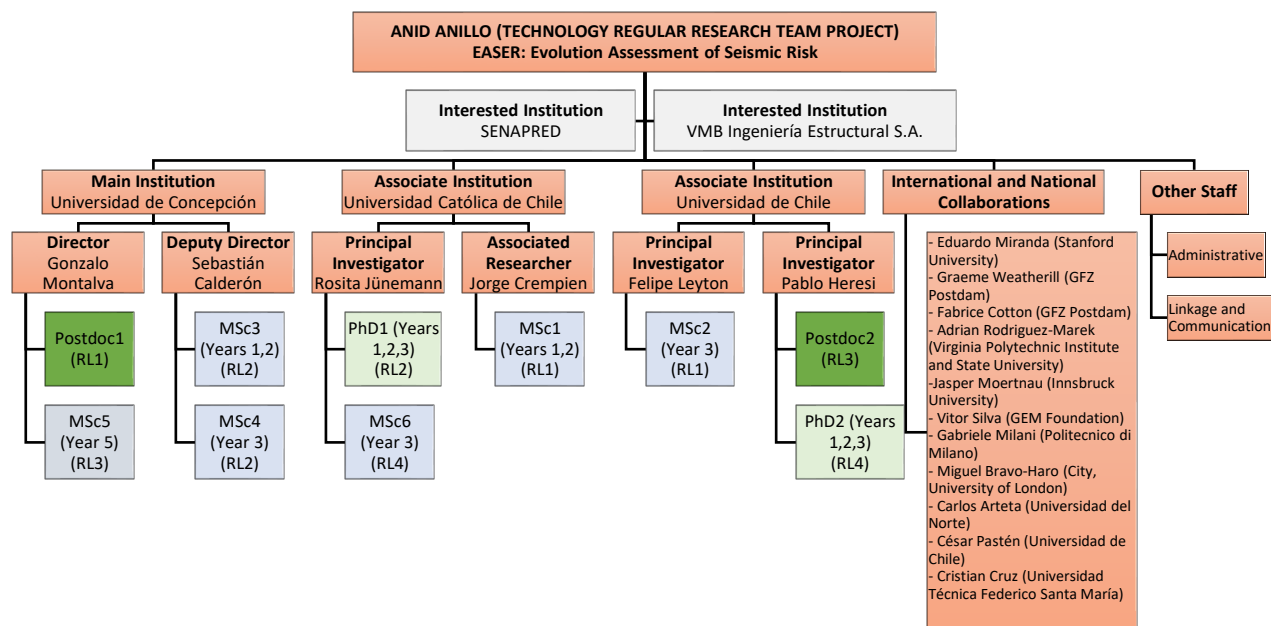


Figure 7. Project organization chart.

VII. PRINCIPAL RESEARCHER RESPONSIBILITIES

Name	Category within the project	Tasks assigned in the Project
Gonzalo Montalva	Director	Project scientific coordination, EASER students' oversight, Co-leading of RL1 and RL3, participation in RL2 and 4. Graduate students advising.
Sebastián Calderón	Deputy Director	Deputy Director, head resource allocation, co-lead RL2 and participation in RL3. Graduate students advising.
Rosita Jünemann	Principal Researcher	Co-lead of RL2 and RL4. Graduate students advising. Gender strategy lead.
Felipe Leyton	Principal Researcher	Co-lead RL1, collaboration in RL4, and students' oversight. Graduate students advising.
Pablo Heresi	Principal Researcher	Co-leading RL3 and RL4, lead the assembly of the computer codes for risk assessment from RL's 1, 2, and 3. Graduate students advising.

VIII. ASSOCIATE RESEARCHER RESPONSIBILITIES

Name	Institutional affiliation	Period of Participation (year)	Research Activities Associated to the Project (Justify His/Her Project Participation)
Jorge Crempien	Pontificia Universidad Católica de Chile	1, 2, 3	Graduate students and Postdoc advising, research within framework of RL1. I will focus on seismicity rates from relocated seismicity catalogs, application of generalized inversion technique to strong motion data to find source, path and site properties, and proposition of megathrust earthquake source properties consistent with time dependent superficial deformation rate data. I will also focus on seismicity rates from relocated seismicity catalogs.

IX. TEAM COLLABORATION

Previous collaborations between other members of the team (last 5 years)		
Collaboration type (article, thesis, project, patent application, congress presentation, etc.)	Title of the collaboration	Team members participating
Seismic Code for Chilean practice	Site-specific Seismic Demand Minimum Calculation Requirements	Gonzalo Montalva (PI), Felipe Leyton (PI), Pablo Heresi (PI), Jorge Crempien (AR), Ian Watt (from VMB).
Seismic Code for Chilean practice	Surface wave methods for estimation of Vs30 and To	Gonzalo Montalva (PI), Felipe Leyton (PI)
Research Project	FONDEF ID22I10032: Obtención de parámetro Vs y T0 basado en técnicas de ondas de	Gonzalo Montalva (PI), Felipe Leyton (PI)
Research Project	FONDEF IT21I0049: Catastrophic Risk Valorization	Rosita Jünemann (PI), Pablo Heresi (PI), Jorge Crempien (AR)
Research Project	FONDEF Plataforma de Amenaza Sísmica SIGAS: Transferencia Tecnológica a Sernageomin y ONEMI mediante estudio de caso en segmento sismo-tectónico de Valparaíso- La Serena	Gonzalo Montalva (PI), Felipe Leyton (PI)
Research Project	EQM160015: Observatory of site response in alluvial basins.	Gonzalo Montalva (PI), Felipe Leyton (PI)
Journal Paper	Strong ground motion prediction model for PGV and spectral velocity for the Chilean subduction zone (2022).	Gonzalo Montalva (PI), Felipe Leyton (PI), Nicolás Bastías (PhD student)
Journal Paper	Measuring coastal subsidence after recent earthquakes in Chile central using SAR interferometry and GNSS data (2023)	Gonzalo Montalva (PI), Felipe Orellana (PostDoc)
Journal Paper	InSAR-Based Early Warning Monitoring Framework to Assess Aquifer Deterioration (2023)	Gonzalo Montalva (PI), Felipe Orellana (PostDoc)
Journal Paper	InSAR Monitoring Using Persistent Scatterer Interferometry (PSI) and Small Baseline Subset (SBAS) Techniques for Ground Deformation Measurement in Metropolitan Area of Concepción, Chile (2023)	Gonzalo Montalva (PI), Felipe Orellana (PostDoc)
Journal Paper	Shallow ambient-noise 3D tomography in the concepción basin, Chile: Implications for low-frequency ground motions (2019)	Gonzalo Montalva (PI), Felipe Leyton (PI)



Journal Paper	Geophysical characterization of the Chilean seismological stations: First results (2018)	Gonzalo Montalva (PI), Felipe Leyton (PI)
Conference Paper	Norma Chilena de Amenaza Sísmica, primeros pasos (2022).	Gonzalo Montalva (PI), Felipe Leyton (PI), Pablo Heresi (PI), Jorge Crempien (AR)
Conference Paper	Avances en la definición de indicadores de incertidumbre en el cálculo del Vs30 basado en métodos de ondas superficiales (2023)	Gonzalo Montalva (PI), Felipe Leyton (PI)
Conference Paper	Cuantificando los Efectos de Sitio empíricos: Primeros resultados del Observatorio de Respuesta de Sitio en Cuencas Aluviales (ORSCA)	Gonzalo Montalva (PI), Nicolás Bastías (PhD student)
Conference Paper	Impacto de la Caracterización Geofísica en la Demanda Sísmica. Giasis Conference (2023)	Gonzalo Montalva (PI), Felipe Leyton (PI)
Conference Paper	Efecto de la Licuación en la demanda Sísmica Giasis Conference (2023).	Gonzalo Montalva (PI), Nicolás Bastías (PhD student)
Conference Paper	Análisis de los coeficientes de correlación espectral de las componentes horizontales y verticales en la subducción chilena. Achisina (2023).	Gonzalo Montalva (PI), Nicolás Bastías (PhD student)
Conference Paper	Variabilidad de resultados de la exploración geofísica de sitios usando métodos de ondas superficiales (2023)	Gonzalo Montalva (PI), Felipe Leyton (PI)
Conference Paper	Evaluación del comportamiento geotécnico-sísmico para puente Milildeo en el terremoto de Melinka (2016, Mw7.6) (2023).	Gonzalo Montalva (PI), Nicolás Bastías (PhD student)
Conference Paper	Regional Dependence of Strong Ground Motion in the Chilean Subduction Zone (2023)	Gonzalo Montalva (PI), Nicolás Bastías (PhD student)
Conference Paper	Modelo Predictivo de Velocidad Máxima (PGV) y Espectral (PSv) para Zona de Subducción Chilena (2018)	Gonzalo Montalva (PI), Felipe Leyton (PI), Nicolás Bastías (PhD student)
Conference Paper	Observatorio de Respuesta de Sitio en Concepción (2018)	Gonzalo Montalva (PI), Felipe Leyton (PI)
Conference Paper	Relación entre períodos altos y Vs: El caso de proyectos de aerogeneradores en Chile (2018).	Gonzalo Montalva (PI), Nicolás Bastías (PhD student)

X. BUDGET AND JUSTIFICATION

This section was filled using the file BUDGET-AND-JUSTIFICATION-2024.xlsx and uploaded in the online application system.

XI. CURRENTLY EXISTING EQUIPMENT

Equipment	Methods	Person in Charge	Location	Date of purchase
ORSCA	Site Response Change Observation	<u>Gonzalo Montalva</u>	<u>Concepción</u>	<u>July 2018</u>
20 ton CPT equipment	Liquefaction Assessment experiment	<u>Gonzalo Montalva</u>	<u>Concepción</u>	<u>May 2021</u>
Mobile Seismometers	System identification of buildings, and site period evolution.	<u>Sebastián Calderón</u>	<u>Concepción-Santiago-Viña del Mar</u>	<u>From 2013 to 2023</u>
Flexible array of 3-component geophones	Characterization of shallow structure using Surface Waves	<u>Felipe Leyton</u>	<u>Santiago</u>	<u>March 2014</u>